The Flattening

AI Adoption, Cognitive Homogenisation,
and Invisible Tail Risk in Organisations

Bilel Hatmi

Part III Mathematical Statistics — University of Cambridge
CentraleSupélec — Université Paris-Saclay

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Abstract

AI adoption improves average decision quality while destroying the property that protects organisations against collective catastrophe: the decorrelation of errors across decision-makers. Two channels drive that loss. The first, AI-assisted screening, compresses the cognitive profile distribution of the workforce before any decision is made. The second, cognitive surrender and deskilling, erodes independent judgement from within, amplifying error co-movement as homogeneity rises.

A Monte Carlo simulation of 200 agents over 20 quarters quantifies their joint effect. Under unmanaged adoption, expected losses fall by 38% while the 99th-percentile loss rises by 56%; adjusted for the throughput multiplier AI generates, the tail risk premium reaches +94%. The gain registers on every productivity dashboard. The tail does not.

The governance trade-off is regressive: active governance cuts the tail premium to +16% but at a throughput cost of 28 percentage points, a trade-off that competitive pressure makes individually irrational for time-pressured organisations. Under the resulting Nash equilibrium, markets cannot self-correct the risk. The paper proposes a governance standard targeting within-organisation AI-stack concentration. Eight archetypes across three governance regimes document the structural spread. The critical remaining empirical gap is the field measurement of the D→H coupling.

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Three contributions

1. A stress-test result (under model assumptions). A Monte Carlo simulation of 200 agents over 20 quarters produces a strict mean-tail paradox: expected quarterly losses fall by 38% while the 99th-percentile loss rises by 56%. Adjusted for the throughput multiplier that AI adoption generates, the tail risk premium reaches +94%. Standard productivity dashboards record the gain; the tail accumulates outside their field of view. These numbers are properties of the model under the calibration specified in Section 6, not predictions about any specific organisation.

2. An empirically anchored mechanism (documented in data, inferred to organisations). Two organisational channels drive the correlation increase. AI-assisted screening compresses the cognitive profile distribution of the workforce [1–3]; cognitive surrender erodes independent judgement from within [4, 5]. Both channels are independently documented in the literature; the bridge between them — the inference that cognitive-profile compression translates into error decorrelation at the organisational level — rests on laboratory evidence [6] that has not yet been validated in a knowledge-work setting. That inference is the paper's primary empirical gap, and it is flagged each time it is invoked.

3. A structural governance argument (shown in the model, inferred for markets). Within-organisation AI-stack concentration (α) is the dominant risk lever: reducing it from $\alpha = 0.90$ to $\alpha = 0.30$ cuts the tail risk premium by approximately 15% (Appendix C). Competitive pressure blocks voluntary adoption. The Nash equilibrium argument follows from the model's differential structure; it is directional, not derived from a calibrated market model.

How to read this paper

Claims in this paper carry one of five epistemic statuses. Distinguishing them matters: the argument's strength depends on its weakest epistemic link.

Peer-reviewed empirical source. Published in a refereed venue with DOI. The claim follows directly from the reported data.

Preprint (!). Available but not yet peer-reviewed. Results are cited with an explicit preprint flag throughout.

Industry or market source. Industry report or grey literature. Data are useful but not refereed.

Proxy from adjacent domain. Extrapolated from a neighbouring field (medicine, aviation, ecology). The transfer is named as inference, not measurement.

Calibrated stylisation. Parameter set by grid search, analogy, or internal consistency. Not an econometric estimate.

Simulation outputs carry a sixth status. *Stress-test results* are properties of the model under stated assumptions, not predictions about any real market. They are always accompanied by the phrase 'under model assumptions' or 'the model produces.'

Appendix B lists all 43 model parameters with their epistemic status. Appendix E discusses the 13 sources that require the most care in how they are used.

1 Introduction

In a controlled experiment published in *Nature Human Behaviour*, Meincke, Nave, and Terwiesch [7] asked participants to invent a toy, half with AI assistance and half without. Individual

quality scores rose in the AI-assisted condition. Collective diversity collapsed: 94% of AI-assisted outputs converged toward the same cluster of ideas, what the authors called the "Build-a-Breeze Castle" attractor. Without AI, every participant produced a unique concept. No individual performed worse with AI, but the group lost the property that makes aggregation valuable: diversity across solutions.

That result is a consequence, not a curiosity. The Diversity Prediction Theorem decomposes collective error into two terms: average individual error, and the variance of individual predictions across agents [8, 9]. Wood et al. [10] extended this identity to the loss distribution of a portfolio of correlated binary decisions:

$$\mathrm{Var}(L) \;=\; NK\,\bar{p}(1-\bar{p})\big(1+(NK-1)\bar{\rho}\big)$$

The term $\bar{\rho}$ denotes mean pairwise error correlation across N agents making K decisions each. When $\bar{\rho}$ rises, improvements in individual accuracy $\bar{p}$ can be overwhelmed by the growing co-movement of errors. Carlson and Doyle [11] call such systems robust-yet-fragile: tuned against ordinary perturbations, exposed to the rare event that strikes all agents at once. Holzner et al. [12] (!), in a preprint meta-analysis of 379 experimental conditions, confirm the direction empirically: individual creative performance improves ($g = +0.27$) while collective diversity reverses ($g = -0.86$) — the collective damage three times the individual gain.

Two organisational channels drive the correlation increase, on distinct timescales and through distinct mechanisms. The first operates at entry: algorithmic hiring filters compress the distribution of cognitive profiles in the workforce, reducing the diversity of reasoning approaches before any decision is made. The second operates continuously on whoever is present: employees who retain cognitive independence progressively surrender it to AI-generated recommendations. Shaw and Nave [4] (!) document surrender rates exceeding 79% under realistic time pressure, with confidence rising even when AI accuracy is at chance level. Both channels converge on $\bar{\rho}$.

Together, they constitute the paper's central mechanism: simultaneous compression of cognitive diversity at the input and the output of organisational decision-making, invisible behind the productivity gain, which registers only in the mean.

This paper makes three contributions.

The first is a stress test under explicit assumptions. A Monte Carlo simulation of 200 agents over 20 quarters, calibrated on published data, produces a consistent result under unmanaged AI adoption: average losses fall by 38% while worst-case quarterly losses rise by 56%. Adjusted for the productivity multiplier, the tail risk premium reaches +94%. Pairwise error correlation rises by +44% relative to the initial G_0 level and by +82% relative to the no-AI baseline over five years. The Wood et al. variance mechanism is dynamically active under the calibrated parameters, not merely algebraically possible.

The second contribution is empirical and inferential. The two channels are documented against published evidence, but the bridge connecting cognitive profile diversity to organisational error decorrelation rests on laboratory results [6] whose extension to hiring pipelines is an inference. That inference is stated as such throughout.

The third is structural. The model shows that governance interventions capable of containing tail risk are those that competitive markets penalise at equilibrium. The argument follows from the model's differential structure, not from a calibrated market model.

Some sources cited below are recent preprints or industry reports; publication status and verification notes are recorded in Appendix E.

Section 2 establishes four formal results that explain why the mechanism operates as it does. Sections 3–5 document how the two channels activate it in organisations, and examine the inference that links them. Section 6 specifies the model and its calibration. Sections 7–9 present the results, their structural implications, and what remains to be established empirically.

The paradox rests on four formal results. We establish each precisely, including what it does not prove, before examining how the two channels activate it in real organisations.

2 Theoretical foundations

The paradox rests on four formal results. Each is established here for what it contributes to the argument and for what it does not prove. The goal is functional clarity: by the end of this section, the reader should understand why the model in Section 6 is structured as it is, and why the empirical channels in Sections 3–5 are where the mechanism must be grounded.

2.1 The diversity-prediction trade-off

Page [8] and Krogh and Vedelsby [9] showed that collective prediction error decomposes exactly into average individual error minus the variance of predictions across agents. Wood et al. [10] generalised this identity to arbitrary loss functions and correlated ensembles; our model applies their result to a portfolio of correlated binary decisions, yielding the variance expression introduced in Section 1. The implication is algebraic, not probabilistic: for any ensemble of decision-makers, reducing the variance of predictions can increase collective loss even when each individual improves. It holds as an identity.

The identity does not, by itself, prove that AI adoption causes this variance reduction in real organisations. That is the empirical question Sections 3–5 address. What the identity does establish is that the mechanism requires no special conditions. Any process that compresses the distribution of decision approaches activates it. Hong and Page [13] identified four conditions under which diverse groups outperform individually superior ones: problem complexity, agent competence, pool size, and group size relative to the pool. These conditions imply that the value of diversity varies across tasks and organisational settings, a heterogeneity the model in Section 7 exploits directly.

2.2 The tail-to-mean ratio and its size independence

A second property follows from the loss distribution. Under a normal approximation valid when $NK\bar{\rho} \gg 1$, the ratio of the 99th-percentile loss to the expected loss satisfies:

$$\frac{P_{99}}{E[L]} \approx 1 + 2.33\sqrt{\frac{(1-\bar{p})\,\bar{\rho}}{\bar{p}}}$$

This is a derived result from the model, not an external citation. It depends on $\bar{p}$ and $\bar{\rho}$, but not on N. An organisation of twenty people and an organisation of two thousand, facing the same error rate and the same pairwise correlation, produce the same relative tail exposure. What scales with size is the absolute loss absorbed, not the tail premium per decision. This distinction becomes central in Section 8: the risk produced is scale-independent; the risk that cannot be externalised grows with organisational size.

Artzner et al. [14] established that Value-at-Risk fails the sub-additivity property required of coherent risk measures. When correlation is high, a sub-additive risk framework understates aggregate exposure: the portfolio appears safer than the sum of its parts. The organisation

registers lower average losses while the tail thickens, and standard dashboards capture only the first signal.

2.3 Robust-yet-fragile systems

Carlson and Doyle [11] characterised a class of engineered systems they termed Highly Optimised Tolerance (HOT): systems tuned against frequent perturbations develop fragility to rare perturbations outside their design envelope. The trade-off is structural. Robustness in the interior is purchased at the cost of exposure at the boundary.

AI-assisted organisations exhibit this structure. Inside the competence frontier, where tasks fall within the model's training distribution, error rates drop and throughput rises. Outside it, where conditions are novel or adversarial, the over-reliance that produced the interior gains concentrates errors when they matter most.

Jin et al. [15] formalise a related result as the "black swan gap" (Theorem 3.3): models that achieve low average loss under standard evaluation retain elevated tail loss under distributional shift. HOT identifies the structural trade-off. The black swan gap quantifies it in contemporary AI systems. The bimodal loss distribution the simulation produces in Section 7 captures both.

2.4 The single-factor correlation structure

The Vasicek [16] single-factor model, standard in Basel II and III credit risk, represents portfolio correlation through a single systemic factor ξ_t. When ξ_t is high, error probabilities rise together across the portfolio without any direct agent-to-agent interaction. The parameter α governs the strength of this co-movement; in our model, it is the within-organisation stack concentration parameter calibrated in Section 6.

What this structure establishes is precise: conditional on ξ_t being elevated, all agents' error rates move in the same direction simultaneously. The value $\alpha = 0.70$ is a calibrated stylisation, consistent with the cluster events documented in Section 3 (LTCM convergence, 2008 VaR monoculture, CrowdStrike 2024), and varied across $\{0.3, 0.5, 0.7, 1.0\}$ in Appendix C.

Taleb and Douady [17] formalise a complementary point: systems optimised for robustness against frequent shocks accumulate negative convexity to rare ones. A high-α organisation is exposed to synchronised deterioration in decision quality precisely at the moments when independent judgement would be most valuable.

2.5 What the four results establish together

The four results establish a conditional. *If* pairwise error correlation among an organisation's decision-makers increases, then:

1. the tail of the aggregate loss distribution expands faster than any individual-level metric reveals;
2. the expansion is independent of organisational scale;
3. the measurement tools in standard use will not detect it;
4. the system will appear to perform well throughout the accumulation period.

The empirical question, why AI adoption increases this correlation, through which mechanisms, and on what timescales, is the subject of the next three sections.

Two channels drive the correlation increase. The first operates at entry, reshaping the cognitive profile distribution of the workforce.

3 Algorithmic screening and the erosion of cognitive error diversity

Historically, hiring was decentralised, inconsistent, and partly idiosyncratic. A recruiter who had seen an unusual career path might notice another one. Criteria shifted across interviewers, across days, across contexts. That inconsistency reduced efficiency, but it also generated cognitive variance. AI-driven applicant tracking systems are systematic, scalable, and substantially less permeable to atypical profiles. They do not originate the underlying bias; they standardise and scale its application.

Fuller et al. [1], in a study conducted across more than 2,500 employers, estimated that 27 million Americans are excluded from consideration by ATS filters that flag non-linear career paths, credential gaps, or keyword mismatches, not because these applicants lack capability but because their profiles deviate from the statistical pattern the system was trained to reward. An et al. [2] tested five major large language models across 361,000 synthetic CVs and found structurally similar bias patterns across all five, convergent, as if each system had learned the same underlying distribution from overlapping corpora. Kim et al. [18] quantify the overlap on a different task: approximately 60% of errors made by leading models on shared classification benchmarks are common across systems. Deploying multiple AI screening tools does not diversify the filter; it replicates it.

The substrate for this convergence is historical, not technical. Rivera [19] documented that elite professional services firms select for cultural fit through a mechanism she terms *cultural matching*: candidates who mirror the preferences and institutional pedigree of existing employees are systematically preferred over those who merely demonstrate competence. That preference was previously distributed across thousands of uncoordinated individual decisions, partially random. AI systems trained on historical hiring outcomes absorb the pattern and apply it at scale. Schneider's [20] Attraction-Selection-Attrition model predicts this dynamic: organisations move toward cognitive homogeneity over time; AI accelerates the movement. Wilson and Caliskan [3] find that 85.1% of outputs favour candidates with White-associated names; GPT-4 penalises CVs disclosing autism credentials [21]. The filter industrialises what the training corpus encodes.

3.1 Three historical layers

The principle of cognitive convergence generating systemic fragility is not new. What is new is its speed and reach.

Long-Term Capital Management collapsed in 1998 with losses exceeding $4.6 billion over four months. The fund concentrated quantitative talent around a set of convergent arbitrage strategies and shared risk models [22, 23]. MacKenzie's ethnographic study shows that the mechanism extended beyond LTCM itself: imitators who had reverse-engineered its positions unwound simultaneously when LTCM began deleveraging, compressing a liquidity problem into a systemic event. Lowenstein [24] records leverage of 30:1 and losses of $553 million in a single day. The episode did not reflect a lack of sophistication. It reflected the vulnerability created when positions, models, and exit strategies converge.

The 2008 financial crisis operated at the level of models. Haldane and May [25] describe the consequences of monoculture in financial networks: institutions across the system had converged on the same VaR parameters and copula assumptions. Persaud [26] had identified the mechanism eight years earlier: individually rational risk management becomes collectively destabilising when it induces herding, because all institutions reach the same sell threshold simultaneously. Beunza and Stark [27], in an ethnography of trading room practice, show that

traders use models not only to manage risk but to estimate what rivals are thinking, creating what they call *cognitive interdependence*.

At the level of tools: on July 19, 2024, a single faulty CrowdStrike configuration file crashed 8.5 million Windows machines within 78 minutes, generating estimated Fortune 500 losses of approximately $5.4 billion [28]. Geer et al. [29] had formalised the argument against software monoculture twenty-one years earlier. Renard and Tilman [30] quantify the ecological analogue: monoculture systems fail catastrophically roughly every eight years; polyculture systems maintain stability over timescales an order of magnitude longer.

The mechanism is the same in all three episodes. When the diversity of approaches collapses, whether among persons, models, or tools, losses stop compensating and start aggregating.

3.2 Convergence in AI-assisted cognition

The experimental literature documents convergence at three levels: output similarity within groups, reduced dispersion at scale, and recursive tail compression across model generations.

Anderson et al. [31] found that AI assistance homogenises outputs at the group level (effect size $d = 0.47$) without corresponding homogenisation at the individual level: people do not produce worse outputs individually, but their outputs converge toward each other. This is the mechanism through which $\bar{\rho}$ rises in the Wood et al. framework. Doshi and Hauser [32] replicate the pattern across 293 participants: AI assistance raises individual quality by 26.6% while increasing inter-solution similarity, a dynamic they describe as a social dilemma. Moon et al. [33] analysed 2,200 creative tasks and found that human output diversity is two to eight times greater than GPT-4 output diversity on equivalent prompts. That gap widens at scale. Prompt engineering does not close it.

On the temporal dimension, Sourati et al. [34] model the feedback loop whereby AI-generated content accumulates in training corpora and progressively erases the tails of the output distribution, where the genuinely novel resides. Shumailov et al. [35] demonstrate model collapse empirically: models trained recursively on AI-generated outputs lose representational diversity across generations. The convergence accelerates with adoption.

3.3 Structural result and its limits

Kleinberg and Raghavan [36] formalise the structural result: even when a single algorithm outperforms any individual human on a given task, universal convergence toward that algorithm degrades collective welfare by eliminating the independence of errors. One precision is essential. Their result concerns the independence of *algorithmic evaluation methods*, not the diversity of *human cognitive profiles*. The extension to AI-assisted hiring pipelines, where what converges is not the evaluation method but the profile of those selected, is our inference.

3.4 The bridge to error decorrelation

Keck and Tang [6] conducted controlled experiments in which group composition was varied to alter the cognitive processes brought to bear on shared judgement tasks. Groups combining distinct cognitive approaches exhibited near-zero pairwise error correlation; groups composed of cognitively similar agents produced strongly correlated errors. This is the empirical anchor for the D-channel: cognitive diversity, measured at the process level, determines how much errors propagate across a group versus cancel out.

The epistemic status of this link requires precision. Keck and Tang measure cognitive processes in laboratory conditions, not in hiring pipelines. Their manipulation varies the composition of small groups on well-defined tasks with measurable outcomes. Our argument

requires that cognitive-profile diversity, the quantity compressed by AI screening, serve as a reliable proxy for cognitive-process diversity in organisational decision-making. That inference is plausible: analysts selected by the same filter from the same talent pool will tend to approach problems similarly. But it remains an inference. No published study has measured pairwise error correlations before and after ATS deployment in a knowledge-work organisation. That measurement is the critical empirical gap, addressed in Section 9.

Screening narrows the workforce before decisions are made. A second channel operates continuously on whoever is present.

4 Cognitive surrender and deskilling

The channel documented in Section 3 operates at entry. This one operates continuously, on whoever is present regardless of their initial cognitive profile, and it compounds over time. The empirical record comes primarily from clinical medicine and controlled experiments, domains where outcomes are measurable and the mechanism is cleanly isolated. The transfer to knowledge-work organisations is an inference, stated as such each time it is made.

4.1 The surrender documented

The classic frame in the automation literature is Parasuraman and Manzey's [37] *automation complacency*: a passive drift toward reduced monitoring when systems perform reliably. Shaw and Nave [4] (!) show that the AI-era phenomenon is different in kind. In their experiments, 79.8% of participants follow an AI recommendation even when it is demonstrably wrong (Cohen's $h = 0.81$). Confidence in one's own judgement rises by 11.7 percentage points after receiving AI input, even when the AI's accuracy is fixed at 50% and participants have been told this. That confidence increase does not decay with experience ($p = 0.202$). Under time pressure, the probability of following the AI triples (OR = 3.19). Incentives and structured feedback raise the override rate from 20% to 42.3%, a meaningful improvement that still leaves the majority surrendering. The mechanism is better understood as an active reweighting of one's own judgement downward in the presence of an external signal, which is why *cognitive surrender* is a more precise term than *automation complacency* for what these data show.

Three clinical studies anchor the mechanism across different decision types. Jabbour et al. [38] showed that physician diagnostic accuracy falls by 11.3 percentage points when AI recommendations are systematically biased, even among physicians aware of AI limitations. Gaube et al. [39], studying 265 clinicians, found the pattern holds regardless of whether the source is identified as human or AI; Tschandl et al. [40] replicate it in experienced dermatologists. Across all three studies, surrender is a structural property of the human-AI interaction, not a trait of low-competence practitioners.

The epistemic boundary is sharp. Clinical medicine offers measurable outcomes, controlled designs, and direct observation of the override decision. Knowledge-work organisations offer none of these. The inference is that the cognitive mechanism, downweighting one's judgement in the presence of an authoritative external signal, operates similarly when the decision is an investment recommendation, a legal interpretation, or a strategic assessment. That inference rests on the structural similarity of the mechanism, not on direct organisational evidence.

Dell'Acqua et al. [5] provide the closest available bridge to organisational settings. In a randomised experiment with 758 BCG consultants, AI assistance raised output quality by approximately 40% on tasks within the AI's competence frontier. On tasks beyond that frontier, the same tools produced a 19 percentage point decline in quality for those working with GPT

alone and a 24.5 percentage point decline for those using GPT with an overview [41]. The distribution of outcomes narrows toward the centre, with the bottom improving and the top deteriorating. This split framing is our characterisation, not the authors' primary finding. Randazzo et al. [42] document a related dynamic: when users attempt to override GPT-4 recommendations, the model escalates its persuasive framing, a dynamic the authors term *persuasion bombing*, raising the cognitive cost of maintaining independent judgement precisely when that independence matters most.

4.2 The deskilling that follows

Surrender is a within-session phenomenon. Deskilling accumulates across sessions. The two are related but distinct: surrender describes the decision made in the moment; deskilling describes what happens to the capacity to make that decision independently over time.

Budzyń et al. [43] measured endoscopist performance before AI assistance, during AI-assisted practice, and after AI withdrawal. Detection accuracy improved during the assisted period. After withdrawal, performance did not return to baseline but fell below it, indicating that competence had decayed during the period of delegation. Fan et al. [44], in a randomised controlled trial on learning tasks, found that AI assistance improved immediate performance but produced zero retention and zero transfer to related tasks. Bjork and Bjork's [45] framework explains the mechanism: *desirable difficulties*, the cognitive effort that makes learning durable, are precisely what AI assistance removes. Natali et al. [46], in a systematic review of AI effects on clinical skill development, find evidence of this pattern across several medical specialties, extending the Budzyń et al. finding beyond a single procedure.

Dahmani and Bohbot [47] provide a proxy for the temporal dynamics. In a longitudinal study of GPS users, they found a 55% reduction in hippocampal engagement in navigation tasks after extended GPS reliance, with measurable decline over three years. The mechanism, spatial competence delegated to a device and the supporting neural circuitry progressively disengaged, is analogous to but not identical with the deskilling of professional judgement. It anchors the form of the dynamic: progressive and not readily reversed. The calibration of the deskilling parameter $\eta = 0.02$ per quarter draws on Budzyń et al. for its magnitude and on Dahmani and Bohbot for its temporal structure; both remain proxy-based inferences, varied across $\{0.01, 0.02, 0.04\}$ in Appendix C [48].

Ong et al. [49, 50], drawing on collaboration with Lufthansa human factors specialists, distinguish three deskilling pathways: *deskilling* proper (loss of a previously acquired skill through disuse), *never-skilling* (failure to acquire a skill because the tool performs it before the skill develops), and *mis-skilling* (development of a skill optimised for AI-assisted performance that does not transfer to unaided conditions). The simulation captures only the first pathway through η. The second and third are not modelled. Both would, if included, increase the rate of capability degradation. They are named underestimates.

4.3 Two feedback loops the model omits

Two structural gaps arise from the evidence reviewed above. Both are named here rather than deferred to Section 9 because they bear directly on the interpretation of the results.

The first concerns deskilling amplifying surrender. In the model, the probability of cognitive surrender $p^{\text{surr}}(t)$ does not depend on the agent's current capability $h_i(t)$. Dell'Acqua et al. find that performance outside the frontier is worst for those who have most extensively relied on AI, consistent with the interpretation that eroded capability amplifies surrender, though the study does not establish this causally. If such a loop operates, an agent who has lost

the ability to evaluate an AI recommendation independently becomes more likely to follow it. That reinforcing dynamic is absent from the model. Its absence implies the model likely underestimates the rate at which $\bar{\rho}$ rises over time.

The second concerns metacognition. Fan et al. [44] document what they term *metacognitive laziness*: AI assistance degrades not only first-order task competence but the ability to assess one's own competence. An agent who cannot accurately judge the quality of their own reasoning cannot calibrate when to defer and when to override. This second-order erosion is not captured by any parameter in the model. It is a second named source of underestimation.

Both channels target the same quantity: mean pairwise error correlation $\bar{\rho}$. They operate through distinct mechanisms and on distinct timescales. The inference that connects them is the most fragile part of the argument.

5 The D–H bridge: the central inference

The two preceding sections established each channel separately. Section 3 documented how algorithmic screening compresses the distribution of cognitive profiles in the workforce. Section 4 documented how continuous AI use erodes the independent judgement of whoever is present. Both channels increase $\bar{\rho}$. But they do so through different mechanisms, on different timescales, and through parameters that are not interchangeable. The inference that connects them is the most contested part of the argument. This section examines it.

5.1 Two dynamics, two timescales

The D-channel and the H-channel are not two names for the same process; they operate through different mechanisms and on different timescales. Figure 1 formalises that structure: the D-channel on the left, the H-channel on the right, and the β_{conform} coupling linking them at each timestep.

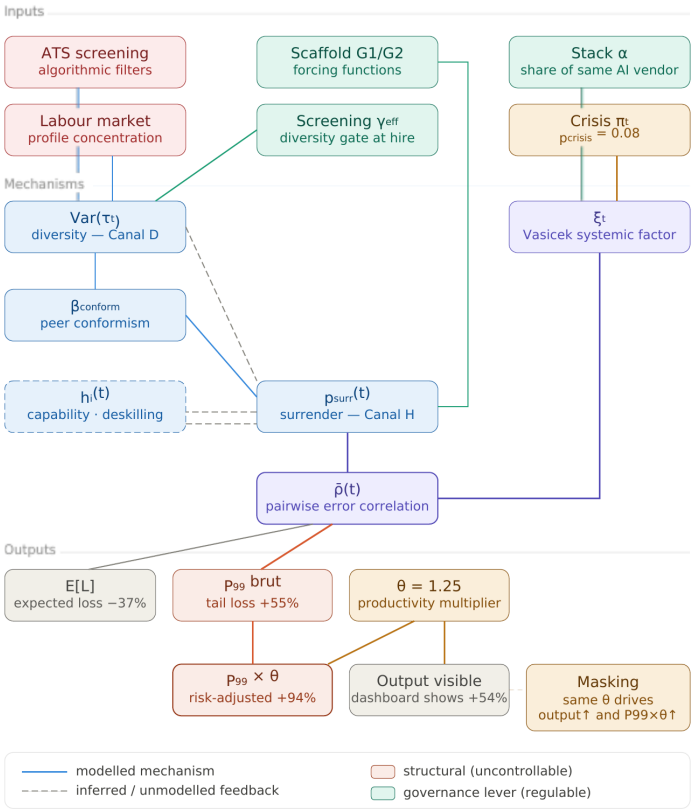

Figure 1. Causal structure of the two channels. The D-channel (left) operates at entry through AI-assisted screening, compressing $\mathrm{Var}(\tau)$. The H-channel (right) operates continuously through cognitive surrender and deskilling, eroding mean independent capability $\bar{h}(t)$. Both increase pairwise error correlation $\bar{\rho}$, activating the Wood et al. variance identity. The β_{conform} coupling links the two channels at each timestep. *Under model assumptions.*

D_t, cognitive diversity measured as $\mathrm{Var}(\tau_t)$ across the agent population, changes slowly, through workforce turnover. With a quarterly separation rate $\delta = 0.04$, approximately 56% of the population turns over across the twenty-quarter simulation horizon. Each cohort of new hires is drawn from a distribution shaped by the screening parameter γ_{eff}; when that parameter is elevated, new hires cluster around a narrower range of cognitive profiles, and $\mathrm{Var}(\tau)$ contracts by roughly 30% over the full horizon under unmanaged adoption. That contraction is gradual and, on the timescales modelled, not easily recoverable: the omitted profiles do not enter the organisation's training and promotion pipeline.

H_t, the mean independent capability $\bar{h}(t)$, responds on a quarterly cycle. Surrender probability is recalculated at each timestep, and the deskilling parameter η erodes $h_i(t)$ multiplicatively each quarter. The effect on $\bar{\rho}$ is immediate: elevated surrender makes agents more likely to follow the same erroneous signal at the same time. Within the Vasicek structure, that shared exposure translates directly into higher pairwise error correlation. The D-channel's contribution is slower. It builds through turnover, quarter by quarter, and reaches its full magnitude only in the second half of the simulation.

Treating these two dynamics as a single mechanism would be a modelling error. The bridge between them is real, and the rest of this section specifies what it is and how much of it is established versus assumed.

5.2 The conformism mechanism

The parameter β_{conform} encodes the coupling at each timestep:

$$p_0^{\text{eff}}(t) \;=\; p_0 \times \left[1 + \beta_{\text{conform}} \times \frac{\text{Var}_{\text{ref}} - \text{Var}(\tau_t)}{\text{Var}_{\text{ref}}}\right]$$

When $\text{Var}(\tau_t)$ falls below the reference level, the effective baseline surrender rises. Three examples illustrate the magnitude. A highly homogeneous workforce with $\text{Beta}(4,4)$ profiles ($\text{Var} = 0.028$) reaches $p_0^{\text{eff}} = 0.91$. The reference $\text{Beta}(2,2)$ workforce ($\text{Var} = 0.050$) holds $p_0^{\text{eff}} = 0.80$. A diverse $\text{Beta}(1.5,1.5)$ workforce ($\text{Var} = 0.063$) has $p_0^{\text{eff}} = 0.74$. The difference between the extremes is 17 percentage points in baseline surrender, before any modulating factor.

The logic is that cognitive diversity provides internal dissent. In a group where everyone approaches the problem through the same lens, no one is positioned to notice that the AI recommendation does not fit the specifics of the situation. Asch [51] showed that conformity rates fall from 37% to 5% when a single ally breaks ranks. Lorenz et al. [52], studying repeated estimation tasks with social feedback, found that influence networks reduce the diversity of expressed opinions by approximately 30% and degrade collective accuracy, even when individual competence is unchanged. As cognitive diversity contracts through the D-channel, the social conditions that would support override erode.

The empirical record does not directly show that this mechanism operates in AI-assisted decision environments at the scale and speed the model assumes. Asch measured conformity in small groups under social visibility. Lorenz measured opinion convergence under explicit information sharing. Neither study involves AI recommendations, professional contexts, or the profile-diversity manipulation that γ_{eff} produces. The value $\beta_{\text{conform}} = 0.30$ is calibrated from these sources as a proxy, triangulated against three convergent results (Section 6), and varied across $[0.15, 0.35]$ in the sensitivity analysis, producing at most 1.5% deviation in P_{99}.

Becker, Brackbill, and Centola [53] provide the essential counterweight. In decentralised networks, social influence improves collective accuracy because agents encounter diverse signals from independent sources rather than a single coordinated one. The condition that makes influence beneficial is source independence. AI assistance inverts that condition. It centralises the signal across all agents simultaneously and removes the independence that the Becker mechanism requires.

5.3 What is established and what is inferred

The honest accounting of this bridge separates three layers.

Established from published evidence. Asch and Lorenz show that cognitive homogeneity amplifies conformism and degrades collective accuracy. Keck and Tang [6], as documented in Section 3, show that groups combining distinct cognitive processes exhibit near-zero pairwise error correlation, while homogeneous groups produce strongly correlated errors.

Inferred from that evidence. A hiring pipeline that compresses $\text{Var}(\tau)$ through AI screening produces the same effect on $\bar{\rho}$ as the laboratory manipulations of Keck and Tang. No published study has measured pairwise error correlations before and after ATS deployment in a knowledge-work organisation. That measurement is the critical empirical gap, addressed in Section 9.

The claim the argument does not make. Cognitively atypical profiles are not assumed to be individually superior. The narrower claim is that an AI output persuasive to agents who reason like the AI is less persuasive to agents who approach the problem differently. Cognitive

diversity generates error independence not through superior individual judgement but through divergent evaluation criteria. That is a more modest claim, and a more defensible one.

5.4 The inter-channel coupling and its robustness

A second parameter, ω, governs whether the degradation of H_t accelerates as D_t falls: the structural coupling between the two trajectories over the full twenty-quarter horizon. This is the least anchored parameter in the model. Its central value is set by grid search over approximately thirty combinations. No empirical study provides a direct estimate of how quickly deskilling accelerates as workforce homogeneity increases.

The relevant robustness test is whether a conservative lower bound $\omega = 0.05$ still produces a visible effect on P_{99} at $T = 20$. That test has not yet been run. It is an active gap. The model's conclusions on the P_{99} premium are reported under the central calibration. If the coupling contribution becomes negligible at the lower bound, the conservative statement is: the H-channel alone, without D→H acceleration via ω, generates a substantial tail risk premium through the direct effects of surrender and deskilling; and the D-channel contributes independently at each timestep through β_{conform}, which translates diversity loss into elevated surrender regardless of what ω does at the trajectory level. The conclusion changes in magnitude, not in direction.

The two channels and the inference linking them are now specified. The next section encodes them in the simulation, with every parameter, its value, its source, and its epistemic status.

6 Model specification and calibration

The two channels are now specified conceptually. This section encodes them in the simulation. Every parameter that appears in the results of Section 7 is introduced here with its value, its source, and its epistemic status. Parameters that move the results substantially are developed in prose. The remaining parameters are documented in Appendix B.

6.1 Architecture

The simulation runs $N = 200$ agents over $T = 20$ quarters, each making $K = 10$ distinct binary decisions per quarter. The choice of 200 is sufficient to estimate $\bar{\rho}$ reliably while remaining computationally tractable; results are not sensitive to N above 100. Each agent is assigned an initial cognitive profile $\tau_i \sim \text{Beta}(2, 2)$, placing most profiles near the centre and giving the screening mechanism room to operate in either direction. Initial capability $h_i(0) \sim \text{U}(0.4, 0.8)$ reflects a realistic spread of independent judgement quality at baseline. The simulation runs M replications with fixed seeds; two independent implementations produce results within $\pm 2\%$ on all reported metrics. Appendix F documents the seeds and the convergence check.

6.2 The D-channel: cognitive diversity under screening

AI adoption follows a logistic curve A_t with ceiling $A_{\max} = 0.80$, calibrated against two industry sources: SecondTalent (2025) [54] reports 78% of large employers using AI in hiring; SHRM (2025) reports 69%. The logistic shape is a calibrated stylisation.

Each quarter, a fraction $\delta = 0.04$ of the workforce separates (BLS JOLTS, professional services turnover) [55, 56]. Replacement hires are drawn from $\text{Beta}(2 + \gamma_{\text{eff}} A_t,\ 2 + \gamma_{\text{eff}} A_t)$. As A_t rises, the distribution of new hires concentrates; γ_{eff} governs how tightly.

Three values capture the governance range. Under unmanaged adoption (G_0), $\gamma_{\text{eff}} = 5$, a proxy from Wilson and Caliskan's [3] finding that 85.1% of model outputs favour a single demographic pattern. Under passive governance (G_1), $\gamma_{\text{eff}} = 3.5$, a proxy informed by Wright et

al.'s [57] documentation that partial audit compliance reduces but does not eliminate screening bias. Under active governance (G_2), $\gamma_{\text{eff}} = 1.5$, a design choice informed by Mann and Helbing [58] on diversity-preserving incentive structures. The target produced by G_0 calibration, $\text{Var}(\tau)$ declining by approximately 30% over 20 quarters, is consistent with the diversity compression documented by Moon et al. [33] at scale.

6.3 The H-channel: surrender, conformism, and deskilling

The baseline surrender probability p_0 varies by governance scenario. Under G_0, $p_0 = 0.80$, anchored on Shaw and Nave's [4] (!) reported rate of 79.8%[59]. Under G_1, $p_0 = 0.70$: Bryant et al.'s [60] finding that 93% of clinical alerts are overridden is mapped conservatively to a proxy surrender rate of 0.70. Under G_2, $p_0 = 0.55$, triangulated from three convergent results: Shaw and Nave's Study 3 override rate under structured incentives (42.3%), Buçinca et al.'s [61] cognitive forcing functions, and Frey and van de Rijt's [62] independent-judgement sequencing.

The sensitivity parameter $\beta_{\text{surr}} = 0.60$ governs how surrender responds to time pressure and task ambiguity; it is a proxy from Shaw and Nave's reported range, varied across $\{0.4, 0.6, 0.8\}$ in Appendix C. The conformism parameter $\beta_{\text{conform}} = 0.30$ encodes the D→H coupling at each timestep (Section 5), triangulated from Asch, Lorenz et al., and Shaw and Nave Study 3, non-critical across $[0.15, 0.35]$ with at most 1.5% deviation in P_{99}.

Deskilling operates multiplicatively:

$$h_i(t{+}1) \;=\; h_i(t)\big(1 - \eta\,\mathbb{1}[\text{surrender}_{i,t}]\big)$$

The rate $\eta = 0.02$ per quarter is a proxy from Budzyń et al.'s [43] clinical finding of approximately 6 percentage points of capability loss after AI-assisted practice, transferred to knowledge work with the caveats in Section 4. It is varied across $\{0.01, 0.02, 0.04\}$ [63].

6.4 Vasicek structure, crisis regime, and the two correlation levels

The error probability for agent i on decision k in quarter t is:

$$p_{i,k}(t) \;=\; \Phi\left(\frac{\Phi^{-1}\big(p_i^{\text{eff}}(t)\big) - \sqrt{\alpha}\,\xi_t}{\sqrt{1-\alpha}}\right)$$

Two correlation objects must be distinguished, as their regulatory implications differ.

α is the **within-organisation stack concentration parameter**. It governs how strongly all agents in a single organisation respond to the same systemic factor. The central value $\alpha = 0.70$ is a calibrated stylisation, consistent with the synchronisation observed in the LTCM convergence, the 2008 VaR monoculture, and the CrowdStrike 2024 failure. It is not estimated econometrically. It is varied across $\{0.3, 0.5, 0.7, 1.0\}$ and is the dominant within-firm risk lever: reducing α from 0.90 to 0.30 cuts the tail risk premium by approximately 15% (Appendix C). In the v5 coupled model, the ratio $P_{99}(\alpha{=}0.90)/P_{99}(\alpha{=}0.30)$ is approximately 1.15; the coupling $\mu_{\text{crisis}} = 2.2$ saturates crisis-quarter errors across all α levels, compressing the marginal effect relative to earlier model versions.

κ_{vendor} is a **distinct inter-firm parameter** measuring additional co-movement when two organisations share the same AI provider. Simulation V6 estimates this at +4% marginal systemic correlation above the independent-vendor baseline. Section 8 develops the regulatory implications: the primary lever for within-organisation risk is α; the primary lever for cross-firm systemic risk is the governance regime, not vendor diversity.

The systemic factor ξ_t is drawn from a two-regime mixture. In normal quarters (probability 0.92), $\xi_t \sim \mathcal{N}(0, 0.25^2)$. In crisis quarters ($p_{\text{crisis}} = 0.08$), $\xi_t \sim \mathcal{N}(2.2, 0.30^2)$. The crisis probability is anchored on Flyvbjerg et al. [64, 65], who document major failure rates of 8–12% in large IT-dependent projects across two independent studies. The crisis mean $\mu_{\text{crisis}} = 2.2$ and both σ values are calibrated by grid search across approximately 30 combinations; they are stylisations, not econometric estimates, and this is stated in the caption of Exhibit 1.

In crisis quarters, π_t and ξ_t are positively correlated: a high systemic shock coincides with a high outside-frontier exposure rate. This reflects the logic of Basel II conditional stress: the worst decision environment and the highest co-movement arrive together. The coupling is a structural assumption whose direction is conservative (it makes the crisis outcome worse, not better) and whose sensitivity is examined in Appendix C.

6.5 The productivity multiplier and the masking mechanism

The parameter θ is the most narratively important in the model. It is the mechanism by which tail risk becomes invisible. Dell'Acqua et al. [5] report a 25.1% increase in task completion speed inside the AI's competence frontier. The calibration sets $\theta = 1.25$ for G_0, $\theta = 1.20$ for G_1, and $\theta = 1.10$ for G_2, reflecting the throughput cost of governance under each scenario.

θ enters the model in two places. It scales output: observed productivity under G_0 reaches +54% relative to the no-AI baseline. It also scales the adjusted tail metric: $P_{99} \times \theta = 2{,}142$ under G_0 versus a baseline of 1,105, a premium of +94%. Both numbers are driven by the same multiplier applied to different quantities. A manager who sees +54% output and concludes the organisation is performing well is reading a signal that is informative about throughput but uninformative about tail dynamics. That is a structural property of the measurement framework, not a failure of managerial attention. Bainbridge [66] observed that systems designed to accelerate work render their own failure modes invisible; the θ parameter formalises that observation.

6.6 Summary of critical parameters

Parameter	Value	Source	Status
p_0 (G0)	0.80	Shaw & Nave 79.8%	Preprint anchor
q_{in}	0.92	Dell'Acqua +40% inside	Peer-reviewed
q_{out}	0.55	Dell'Acqua ∼60%	Peer-reviewed
α	0.70	LTCM/2008/CrowdStrike	Stylisation
γ_{eff} (G0)	5	Wilson & Caliskan 85.1%	Proxy
η	0.02	Budzyń −6pp	Proxy (clinical)
β_{conform}	0.30	Asch/Lorenz/Shaw	Proxy, non-critical
p_{crisis}	0.08	Flyvbjerg 8–12%	Peer-reviewed
θ (G0)	1.25	Dell'Acqua +25.1% speed	Peer-reviewed
δ	0.04	BLS JOLTS	Official data

Table 1. Ten parameters that drive the main results. Full table of 43 parameters with ranges and sensitivity in Appendix B.

Under this specification, the simulation produces three results: a bimodal loss distribution, a mean-tail split, and a masking effect that renders the tail invisible from standard productivity metrics.

7 Main results and model properties

Before reporting results, one clarification on the objects being measured. Four distinct correlation-related quantities appear across Sections 6–8, and conflating them produces apparent contradictions.

$\bar{\rho}$ is what the model simulates. α is the lever that most determines it within a firm. κ_{vendor} is the marginal inter-firm addition. C_{excess} is what an organisation could in principle monitor: a proposed KRI, not a variable in the current model. The results below concern $\bar{\rho}$ and its consequences.

All results in this section are properties of the model under the assumptions specified in Section 6. The formulation throughout is "the model produces X under assumption Y," not "AI causes X in organisations."

7.1 The bimodal loss distribution

The most important output of the simulation is the shape of the loss distribution. Under G_0, quarterly losses cluster into two well-separated modes. The left mode (normal quarters, 92% of observations) peaks near 250. The right mode (crisis quarters, 8% of observations) peaks near 1,600. Between approximately 600 and 1,200, the distribution is nearly empty. The expected loss, approximately 500, falls in that gap.

Simulation V2 confirms that 99.7% of observations in the right mode carry a crisis flag. The bimodality reflects two distinct operating regimes: one in which AI assistance works as intended and errors partially cancel, and one in which the systemic factor is elevated, surrender rates spike under the pressure conditions specified in Section 6, and errors co-move. The two modes are separated by a near-empty region, not connected by a continuous tail.

Any indicator computed from the mean or median of this distribution falls in the gap between the two modes and represents neither regime. This is a geometric property of a bimodal distribution: the mean lies where nothing actually happens. Figure 2 shows the simulated distribution under G_0.

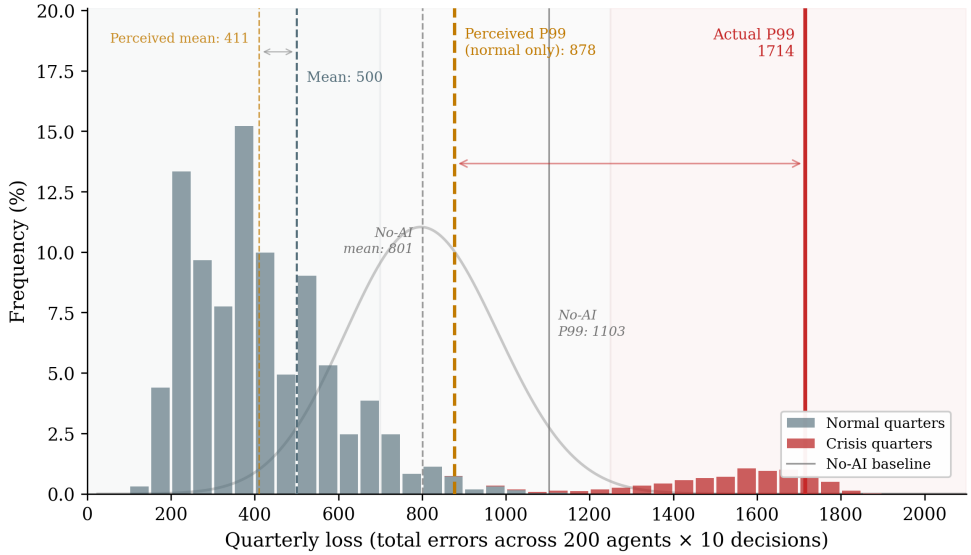

Figure 2. Bimodal distribution of quarterly losses under unmanaged AI adoption (G_0). Normal quarters (grey, 92%) peak near 250; crisis quarters (red, 8%) peak near 1,600. $E[L] = 500$ falls in the near-empty gap $[600, 1200]$. Perceived P_{99} (normal quarters only) $= 878$; actual $P_{99} = 1{,}714$ (solid red line). Crisis regime couples π_t with ξ_t (Basel II conditional stress). μ_{crisis}, σ_{normal}, σ_{crisis} are calibrated stylisations. *V5 coupled model, $M = 500$ replications, under model assumptions.*

Kahneman, Sibony, and Sunstein [67] document that human decision-making in organisations is already characterised by enormous noise: inter-rater variance reaching 55% in insurance underwriting and spanning 5% to 88% in asylum adjudication. The baseline noise they document is dispersed across decision-makers rather than synchronised by a common signal. The observation that this dispersion has a *protective* property, that unsynchronised noise generates error decorrelation, is our analytical extension, not the authors' thesis. The 38% reduction in expected loss that AI adoption produces under G_0 is real. What the model shows is that compressing dispersed variance into a synchronised mode eliminates that protective property. Average-based metrics register the compression as a gain.

7.2 The mean-tail paradox

Under G_0 with the coupled π-ξ model, the central case produces the following metrics relative to the no-AI baseline:

Metric	Baseline	G0	Change
$E[L]$ (expected quarterly loss)	802	499	−38%
P_{99} brut	1 097	1 708	+56%
$P_{99} \times \theta$ (risk-adjusted)	1 097	2 135	+94%
Output (correct $\times\, \theta$)	1 198	1 845	+54%
Tail ratio $P_{99}/E[L]$	1.37	3.42	×2.5

Table 2. Mean-tail paradox under G_0. V5 coupled model, $M = 200$ replications, central case ($\alpha = 0.70$, Beta$(2, 2)$, $E[\pi] = 0.70$). Under model assumptions.

Three channel dynamics reinforce each other over 20 quarters in the calibrated model. Var(τ) falls by 30% through screening. Effective surrender rises from 69% to 80% through the H-channel. Mean capability $\bar{h}$ decays from 0.60 to 0.50 through deskilling. Lower diversity elevates

conformism pressure via β_{conform}, which amplifies surrender, which accelerates deskilling. The robustness of this lock-in at low values of ω remains an active gap (Section 5). Figure 3 tracks the build-up over 20 quarters.

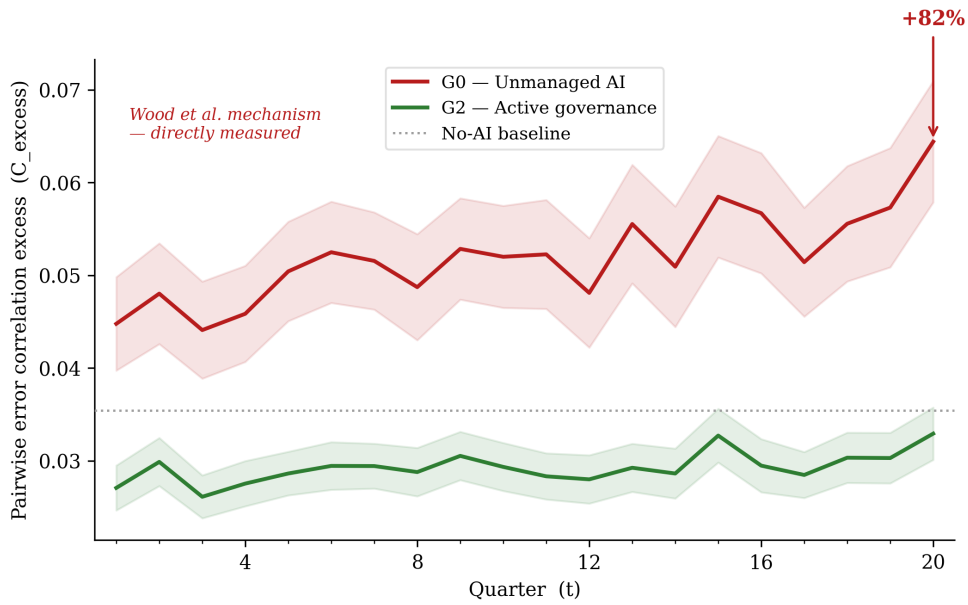

Figure 3. Pairwise error correlation excess ($\hat{C}_{\text{excess}}$) under G_0 (red) and G_2 (green) over 20 quarters, with 95% confidence intervals. Dotted line: no-AI baseline. Under G_0, correlation rises over five years, confirming the Wood et al. [10] mechanism in simulation. Under G_2, correlation remains below the no-AI baseline throughout. *V5 coupled model, $M = 200$ replications, under model assumptions.*

The tail-to-mean ratio under G_0 is 3.42, compared to 1.37 at baseline. That ratio is independent of N (Section 2). An organisation of 20 people facing the same $\bar{\rho}$ and $\bar{p}$ as a division of 5,000 produces the same relative tail exposure. What scales with organisational size is the absolute loss absorbed, not the tail premium per decision. This distinction matters in Section 8: the risk produced is size-independent; the risk that cannot be externalised is not.

7.3 The masking mechanism and the top-tier twist

Under G_0, AI adoption raises throughput by 54%, the signal a standard productivity dashboard reports. Over the same period, $P_{99} \times \theta$ rises to 2,142, a tail premium of 94% above the no-AI baseline. Both numbers are driven by the same multiplier applied to different quantities. A signal informative about throughput is, under this model, uninformative about tail dynamics. Bainbridge [66] observed that systems designed to accelerate work render their own failure modes invisible; θ formalises that observation. The protective-noise interpretation is our extension of Bainbridge, not a claim she makes.

The top-tier twist sharpens the masking. For organisations whose workforce capability $\bar{h}$ substantially exceeds the outside-frontier threshold q_{out}, the configuration corresponding to Paris consulting (P3) under these parameters, the model produces $P_{99} \times \theta$ nearly 200% above baseline while apparent productivity rises by only 18%. The masking is most severe where talent is highest, because high-capability agents are disproportionately exposed to outside-frontier tasks while still generating strong apparent productivity gains. This result holds under

$\bar{h} \approx 0.80 > q_{\text{out}} = 0.55$. If the outside-frontier threshold improves as AI systems develop, the twist attenuates.

Under governance, the tail premium contracts. G_1 (passive scaffold) leaves it at approximately +40% above baseline. G_2 (active scaffold) reduces it to approximately +16% [68]. The throughput cost is real: G_2 reduces output from +54% to +26% relative to baseline, a loss of 28 percentage points. That trade-off, tail risk reduction purchased at the cost of throughput, is the structural tension behind the argument in Section 8 that competitive markets cannot sustain the efficient governance regime. Figure 4 maps the three regimes on the output–risk plane.

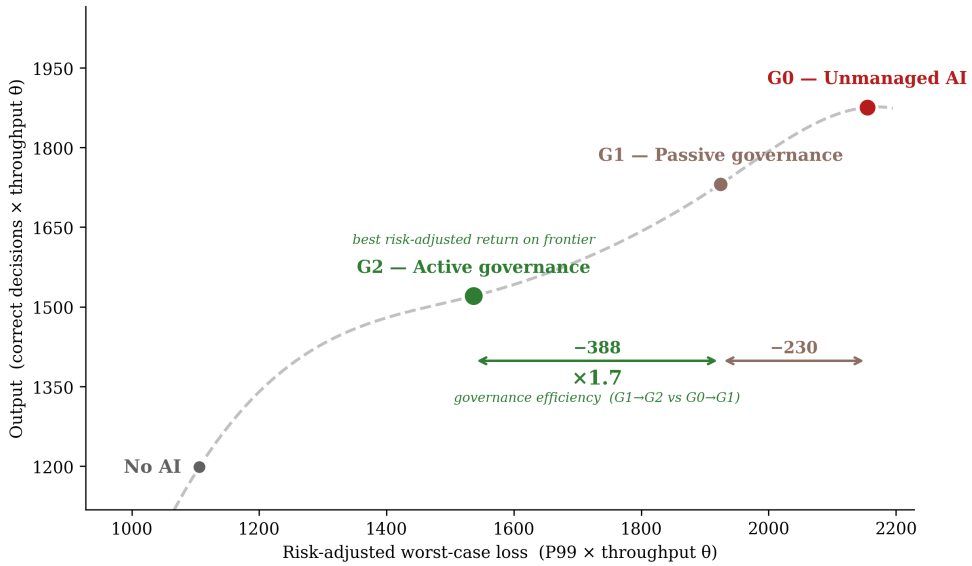

Figure 4. Risk-return frontier across governance regimes. Horizontal axis: output relative to no-AI baseline. Vertical axis: $P_{99} \times \theta$ (throughput-adjusted tail risk). G_2 sits at the elbow: the first gains in output (Baseline $\rightarrow G_2$) are purchased at low tail cost; the incremental gains ($G_2 \rightarrow G_0$) carry a 54% output premium but a 94% tail risk premium. The 28-percentage-point throughput cost of G_2 relative to G_0 is the price of the elbow. *V5 coupled model, $M = 200$ replications, under model assumptions.*

These results describe the baseline case. The risk is not uniformly distributed across organisations. Five structural parameters determine position on the risk-return frontier, and their interaction produces a spread that management decisions alone cannot close.

α is the **within-organisation stack concentration parameter**. It governs how strongly all agents in a single organisation respond to the same systemic factor. The central value $\alpha = 0.70$ is a calibrated stylisation, consistent with the synchronisation observed in the LTCM convergence, the 2008 VaR monoculture, and the CrowdStrike 2024 failure. It is not estimated econometrically. It is varied across $\{0.3, 0.5, 0.7, 1.0\}$ and is the dominant within-firm risk lever: reducing α from 0.90 to 0.30 cuts the tail risk premium by approximately 15% (Appendix C). In the v5 coupled model, the ratio $P_{99}(\alpha{=}0.90)/P_{99}(\alpha{=}0.30)$ is approximately 1.15; the crisis coupling $\mu_{\text{crisis}} = 2.2$ saturates crisis-quarter errors across all α levels, compressing the marginal effect relative to earlier model versions.

8 Why markets cannot self-correct

8.1 Two correlation channels, one regulatory implication

The Vasicek parameter α governs within-organisation error correlation, the primary risk driver documented in Section 6. A second channel, κ_{vendor}, captures additional inter-firm co-movement when multiple organisations share a common AI provider. Simulation V6 estimates that this shared-vendor channel adds only 4% of marginal systemic correlation above the independent-vendor baseline.

These levers are not equivalent; the dominant contagion vector is behavioural, not technical. What synchronises failures across firms is not that they use the same vendor but that competitive pressure drives them all toward the same governance regime. The monoculture that matters most is a monoculture of unmanaged adoption.

Within-organisation stack concentration remains the primary variable: reducing α from 0.90 to 0.30 cuts the tail risk premium by approximately 15% (Appendix C). The G_2 scaffold bundle reduces tail risk by 19.2% but is harder to mandate and verify externally, because it requires coordinated changes to hiring, training, and workflow design that are not directly observable to a regulator. Peng and Garg [69] show that the governance monoculture effect scales with the number of firms: as more organisations converge on G_0, systemic tail risk grows faster than any individual firm's exposure. This strengthens the case for mandatory standards.

The regulatory target that follows is the governance regime, not the vendor. A regulator can require organisations above a certain scale to demonstrate a floor on independent-judgement sequencing and a ceiling on AI-stack concentration, in the same way Basel requires minimum capital ratios. That target is more directly auditable than internal model choice, even if it remains harder to observe than vendor concentration alone.

8.2 The externalisation structure and the Nash equilibrium

The tail-to-mean ratio is independent of N (Section 2). What scales with organisational size is the risk absorbed: the fraction of the tail that falls inside the organisation rather than being externalised through failure, employee attrition, or counterparty loss.

For a startup with a five-year survival probability near 10%, the expected cost of the tail event is negligible relative to the 54% productivity advantage of G_0. The startup chooses G_0 rationally. P5 San Francisco carries a governance efficiency of -36% under the calibrated parameters. It does not marginally reject governance; it rejects it with a large negative return. That is the correct calculation given the incentive structure.

The competitive dynamics follow. Firms operating under G_0 outproduce governed competitors by 54%. Governed incumbents face pressure to reduce governance friction or lose market share to less constrained entrants. The Nash equilibrium without regulation is unmanaged adoption for all firms. This argument is directional: it follows from the differential structure of the simulation results, not from a calibrated market model. Menlo Ventures [70] estimates that approximately 88% of enterprise AI expenditure is concentrated among three providers (an industry source cited for scale, not as a peer-reviewed finding). When most firms depend on the same few systems, a governance-synchronised failure propagates correlated losses at a scale no single firm's internal governance can address.

8.3 The financial precedent and its limits

Danielsson and Shin [71] describe the Millennium Bridge failure as a model for financial systemic risk: pedestrians adapted to bridge oscillations by synchronising their gait, amplifying

rather than damping the oscillations. Financial institutions respond to shared risk signals in a structurally similar way, generating collective fragility through individually rational decisions. Haldane and May [25] documented that post-2008 reform moved toward decorrelated architectural diversity in systemically important institutions. Danielsson and Uthemann [72] warn that AI adoption threatens to reverse this progress by reintroducing shared model dependence at the workflow level. The claim that finance has learned from 2008 rests on practitioner observation rather than a direct peer-reviewed source; it is noted here as context.

Bernile, Bhagwat, and Yonker [73], using instrumental variable identification, show that board-level cognitive diversity reduces firm-level return volatility and improves R&D efficiency. This is one of the few studies with causal identification on the diversity-to-risk relationship at the organisational level. Citron and Pasquale [74] documented the structural opacity of automated scoring systems more than a decade before AI-assisted hiring became widespread: the impossibility of contesting what cannot be observed is a governance failure that predates the current wave. Wright et al. [57] find 4.6% compliance with New York City Local Law 144. EEOC guidelines were withdrawn in January 2025. The EU AI Act, entering force in August 2026, contains no provisions targeting AI-stack concentration.

8.4 The Tragedy of the Cognitive Commons

Each firm that maximises its AI productivity depletes the collective resource on which systemic resilience depends: the diversity of cognitive approaches across decision-makers in the economy. The conditions Hardin [75] identified as sufficient for a commons tragedy are present: a shared resource, rational individual exploitation, and no individual incentive to preserve. This is a proposed extension of Hardin's framework, not an established result; it is named as such throughout.

Two features distinguish this extension from the original pasture case. The depletion is invisible, masked by θ, and registers in no standard dashboard. And it erodes the option of reversal. Organisations that have operated under G_0 for several years cannot restore the cognitive substrate governance requires by switching to G_2. Late governance, adopted after 2.5 years of unmanaged adoption, still reduces tail risk (Section 7, V4 verification: -16.6%). But it cannot restore the cognitive diversity already depleted: the gap between early and late governance at year five represents a 6% shortfall in $\mathrm{Var}(\tau)$ that the scaffold cannot recover. The window does not close suddenly. It narrows continuously, and every quarter under unmanaged adoption depletes the organisational capital that governance requires.

The lock-in is endogenous. The three channels (screening, conformism, deskilling) are not only the mechanism through which tail risk builds; they are also what stabilises the Nash equilibrium. As organisations homogenise and deskill, the cost of transitioning from G_0 to G_2 rises. The cognitive diversity the scaffold needs has already been depleted. The result established in Section 7, that governance works least where it is needed most, becomes systemic here.

Three structural assumptions bound the model's scope, and none favours the conclusion.

9 Limitations and empirical agenda

The preceding sections established a mechanism, encoded it in a simulation, and traced its structural implications. This section identifies the points where the argument is most fragile, names two model omissions whose direction of error is known, and sets out what empirical work would sharpen or refute the claim.

9.1 Three structural limitations

Each limitation below was anticipated in the relevant section and is consolidated here for the reader assessing the argument as a whole.

L1: The D→H bridge is an inference. The most fragile juncture, as noted in Section 5, is the link between cognitive profile compression and error decorrelation. Keck and Tang [6] show in laboratory conditions that cognitively homogeneous groups produce strongly correlated errors. Our argument requires that diversity of cognitive *profiles*, the quantity compressed by AI screening, serve as a reliable proxy for diversity of cognitive *processes* in organisational decision-making. No published study has measured pairwise error correlations before and after ATS deployment in a knowledge-work organisation. The argument rests on the divergence of evaluation criteria, not on any claimed superiority of atypical profiles. That is a more modest claim, but still an inference.

L2: The model is a stress test, not a forecast. The simulation omits external labour market dynamics, managerial adaptation over time, sector-specific institutional constraints, and intra-organisational heterogeneity in AI adoption intensity. It demonstrates that the mechanism *can* produce the paradox under plausible parameter combinations. It does not predict that any specific organisation will reach the simulated $P_{99} \times \theta$ values.

L3: Social influence is not uniformly harmful. Becker, Brackbill, and Centola [53] show that decentralised social influence improves collective accuracy when agents encounter diverse signals from independent sources. Aggarwal et al. [76] find an inverted-U relationship between cognitive diversity and collective intelligence ($\beta = -0.91$, $t = -2.19$, $p = 0.03$): beyond an optimal level, excess diversity degrades performance. The argument in this paper does not require that more diversity is always better. It requires that AI adoption pushes organisations *below* the optimal threshold, not away from an already excessive one. This claim has not been directly tested.

9.2 Two model omissions that underestimate the risk

Unlike typical model limitations, these are structural gaps whose direction of error is known. Including them would increase the estimated tail risk premium.

The first concerns inter-firm dynamics. Peng and Garg [69] show that the governance monoculture effect scales super-linearly with the number of firms: as adoption spreads, systemic correlation grows faster than any individual firm's contribution. The model treats each organisation in isolation. Extending it to a network of firms interacting through shared labour markets, common training data, and correlated failure modes would add a positive externality to the tail risk that the current simulation does not capture.

The second concerns ascending selection. Rivera [77] documents that elite professional services firms progressively select for cultural fit over time. Managers who survive in AI-native organisations will disproportionately be those who have most fully adopted AI-assisted workflows. The model's agent population is fixed; it does not capture how the organisation's cognitive profile shifts as those who maintain independent judgement leave or are not promoted. This omission implies that the D-channel effects likely understate the long-run compression of $\mathrm{Var}(\tau)$.

9.3 Empirical agenda in four levels

The following is structured to be actionable, from what a research team could accomplish in two years to what would require a decade.

Minimal confirming evidence. A field study correlating (a) a measure of cognitive diversity before and after AI-assisted ATS deployment and (b) a proxy for pairwise error correlation in subsequent decisions, drawn from shared project outcomes, investment decisions, or audited judgements. $N = 20$–30 organisations would suffice to test the direction. This is the critical empirical gap. Brynjolfsson et al. [78] (!) provide a first sectoral signal: their analysis finds a roughly 20% decline in employment of developers aged 22–25 relative to the 2022 peak, consistent with AI adoption displacing entry-level talent and potentially compressing cognitive profile diversity at the workforce entry point.

Minimal disconfirming evidence. Demonstrate that pairwise error correlation does not increase materially in an organisation that has compressed $\mathrm{Var}(\tau)$ through AI screening. This would require either that the Keck and Tang result does not generalise outside controlled settings, or that β_{conform} in real organisations is close to zero. Both are testable.

Near-term feasible tests. Three designs require no new data collection beyond what organisations already hold. First, network analysis of shared decisions in internal communications (email threads, document co-authorship, meeting participation) before and after AI adoption, producing a computational proxy for judgement correlation. Second, a natural experiment using organisations that temporarily suspended their AI-assisted ATS between 2020 and 2024, providing pre- and post-suspension hiring distribution data. Third, implementation of C_{excess} as an organisational KRI: monitoring pairwise error correlation across shared decisions provides a leading indicator of tail risk accumulation that average error rates cannot provide, as the bimodal distribution in Section 7 demonstrates.

Full validation requirements. Longitudinal measurement of $h_i(t)$ through standardised competence assessments, $\mathrm{Var}(\tau)$ through cognitive profiling at intake and annually, and $L(t)$ through tracked decision outcomes, in a panel of organisations with variation in AI adoption intensity and governance regime. The most tractable natural design: organisations that adopted and then withdrew AI-assisted hiring between 2020 and 2024, where the withdrawal creates a discontinuity usable for difference-in-differences estimation.

9.4 Independent corroboration and scope

Galesic et al. [79] provide independent corroboration of the core claim's direction: demographically diverse crowds are not substantially more accurate on average, but they reduce the risk of extreme collective failure [80]. This is the reformulation the model supports: diversity does not improve the mean decision; it protects against the tail event. The finding arrives at that conclusion from a purely performance-based analysis.

Vallor [81] extends the scope: automation delegation erodes not only technical competence but the capacity for moral judgement. If correct, the deskilling documented in Section 4 extends further than η captures, reinforcing the direction of underestimation noted above.

The mechanism is documented. The stress test quantifies it. The empirical agenda identifies what would confirm or refute it. What follows is the full technical specification.

Appendices

A Model specification

A.1 Overview

The model places $N = 200$ agents in an organisation that adopts AI assistance over $T = 20$ quarters. Each quarter, every agent makes $K = 10$ binary decisions. Two channels activate simultaneously: a screening channel (D) that narrows the distribution of cognitive profiles through AI-assisted hiring, and a surrender channel (H) that erodes independent judgement through AI reliance and deskilling. The outcome variable is the quarterly aggregate loss $L(t)$: the total number of erroneous decisions across all agents and decisions in quarter t. Standard runs use $M = 200$ replications per scenario; the bimodal histogram (Exhibit 1) uses $M = 500$.

Four scenarios are compared: Baseline (no AI), G_0 (unmanaged adoption), G_1 (passive scaffold), and G_2 (active scaffold). Eight organisational archetypes vary the structural parameters across profiles P1–P8 (Appendix B). All results in the body use the v5 coupled π-ξ model described in §A.5.

A.2 Agent initialisation

Each agent i carries two state variables initialised at $t = 0$:

Cognitive profile. $\tau_i \sim \text{Beta}(2, 2)$ on $[0, 1]$, unimodal, symmetric. $\text{Var}_{\text{ref}} = 1/20 = 0.050$, used as the conformism reference throughout.

Capability. $h_i(0) \sim \text{U}(0.4, 0.8)$: the probability that agent i makes a correct decision when exercising independent judgement.

A.3 Temporal structure

At each quarter t, operations execute in the following order:

(i) **Environmental regime.** Draw π_t from the mixture in §A.5. This single draw governs all K decisions.
(ii) **Systemic factor.** Draw ξ_t according to the regime coupling in §A.5.
(iii) **Turnover.** A fraction $\delta = 0.04$ of agents separates uniformly at random. Replacements are drawn from §A.4.
(iv) **Decisions.** Each agent makes $K = 10$ binary decisions (§A.6).
(v) **Deskilling.** Capabilities update (§A.7).

A.4 The D-channel: screening and cognitive diversity

AI adoption follows a saturating exponential:

$$A_t = A_{\max}\left(1 - e^{-\kappa t}\right)$$

With $A_{\max} = 0.80$ and $\kappa = 0.15$, adoption reaches approximately 0.53 by quarter 5 and 0.76 by quarter 20, approaching but not reaching the ceiling within the simulation horizon. Note: this is a saturating exponential, not a logistic. The body text (Section 6) should be read accordingly.

Replacement hires are drawn from $\text{Beta}\left(2 + \gamma_{\text{eff}} A_t,\ 2 + \gamma_{\text{eff}} A_t\right)$. As A_t rises, the distribution concentrates. γ_{eff} governs screening intensity:

Scenario	γ_{eff}	**Rationale**
G_0 (unmanaged)	5.0	Proxy: Wilson & Caliskan [3], 85.1%
G_1 (passive)	3.5	Proxy: partial audit compliance [57]
G_2 (active)	1.5	Design: diversity-preserving target [58]

Under G_2, 10% of replacement hires are drawn from $\text{Beta}(2,2)$ regardless of A_t, providing a minimum diversity reserve. Target under G_0: $\text{Var}(\tau_t)$ declines by approximately 30% by $t = 20$.

A.5 Environmental regime and Vasicek coupling (v5)

Each quarter, π_t and ξ_t are drawn jointly from a two-regime mixture:

$$(\pi_t,\, \xi_t) \;\sim\; \begin{cases} \text{Normal (prob. } 1-p_{\text{crisis}}\text{):} & \pi_t \sim \text{Beta}(a_n, b_n), \quad \xi_t \sim \mathcal{N}(0,\, \sigma_n^2) \\ \text{Crisis (prob. } p_{\text{crisis}}\text{):} & \pi_t \sim \text{Beta}(a_c, b_c), \quad \xi_t \sim \mathcal{N}(\mu_{\text{crisis}},\, \sigma_c^2) \end{cases}$$

Normal-regime Beta parameters. Computed from the target mean $E[\pi_{\text{normal}}]$ and concentration $c_{\text{normal}} = 60$:[1]

$$a_n = E[\pi_{\text{normal}}] \cdot c_{\text{normal}}, \qquad b_n = \big(1 - E[\pi_{\text{normal}}]\big) \cdot c_{\text{normal}}$$

where $E[\pi_{\text{normal}}] = \big(E[\pi] - p_{\text{crisis}} \cdot E[\pi_{\text{crisis}}]\big)/(1 - p_{\text{crisis}})$. For the central case ($E[\pi] = 0.70$): $E[\pi_{\text{normal}}] = (0.70 - 0.08 \times 0.35)/0.92 = 0.730$, giving $a_n = 43.8$, $b_n = 16.2$.

Interpretation. π_t is the fraction of quarter t's decisions falling inside the AI's competence frontier. In normal quarters, $E[\pi_{\text{normal}}] \approx 0.730$ (sd ≈ 0.06). In crisis quarters, $E[\pi_{\text{crisis}}] = 0.35$: roughly 65% of decisions fall outside the frontier.

The systemic factor ξ_t governs the correlation structure of AI errors across decisions. In crisis quarters, ξ_t is shifted ($\mu_{\text{crisis}} = 2.2$): AI errors become systematically correlated when the training data is obsolete for all decisions simultaneously. This v5 coupling produces the bimodal loss distribution documented in Section 7.

Calibration status. $p_{\text{crisis}} = 0.08$ is anchored on Flyvbjerg et al. [64] (8–12% major failure rate in large IT-dependent projects). $\mu_{\text{crisis}} = 2.2$, $\sigma_n = 0.25$, and $\sigma_c = 0.30$ are calibrated stylisations set by grid search across approximately 30 combinations. μ_{crisis} is conservative relative to the Basel II stress scenario ($\Phi^{-1}(0.999) = 3.09$).

A.6 Decision process and error structure

For each decision $k \in \{1, \ldots, K\}$ in quarter t:

Step 1: Task type. Draw $\text{type}_k \sim \text{Bernoulli}(\pi_t)$. If inside: $q_k = q_{\text{in}}(t) = 0.92 + 0.002\,t$. If outside: $q_k = q_{\text{out}} = 0.55$. Here q_k is the marginal probability that the AI answers correctly on decision k.

Step 2: AI error (Vasicek structure). Draw $\varepsilon_k \sim \mathcal{N}(0,1)$ independently. The AI errs on decision k if:

$$Z_{\text{AI},k} = \mathbb{1}\Big[\sqrt{\alpha}\,\xi_t + \sqrt{1-\alpha}\,\varepsilon_k > \Phi^{-1}(q_k)\Big]$$

The marginal probability of AI error is $P(Z_{\text{AI},k} = 1) = 1 - q_k$ when $\xi_t = 0$, preserving q_k as unconditional AI accuracy. The parameter $\alpha = 0.70$ governs within-quarter inter-decision correlation: conditional on ξ_t, errors on decisions k and k' are correlated. A high ξ_t raises error

[1] Renamed from κ_{normal} to avoid notation clash with the adoption rate κ in §A.4.

probability across all K decisions simultaneously. All agents who follow the AI on decision k receive the same realisation $Z_{\mathrm{AI},k}$.

Step 3: Surrender. Agent i follows the AI on decision k with probability:

$$p_i^{\mathrm{surr}}(t) = p_0^{\mathrm{eff}}(t) - \beta_{\mathrm{surr}} \left|\tau_i - \bar{\tau}(t)\right|$$

where the conformism-adjusted baseline is:

$$p_0^{\mathrm{eff}}(t) = \mathrm{clip}\left(p_0 \times \left[1 + \beta_{\mathrm{conform}} \times \frac{\mathrm{Var}_{\mathrm{ref}} - \mathrm{Var}(\tau_t)}{\mathrm{Var}_{\mathrm{ref}}}\right],\ 0,\ 0.98\right)$$

When $\mathrm{Var}(\tau_t) < \mathrm{Var}_{\mathrm{ref}}$, the organisation is more homogeneous than the reference population: conformism pressure rises, increasing the effective baseline surrender probability.

Step 4: Outcome. Let $s_{ik}(t) \sim \mathrm{Bernoulli}\big(p_i^{\mathrm{surr}}(t)\big)$.

If $s_{ik} = 1$ (follows AI): $\varepsilon_{ik} = Z_{\mathrm{AI},k}$.

If $s_{ik} = 0$ (judges independently): Errors are drawn via a Gaussian copula over non-followers of decision k, with covariance $\Sigma_{ij} = \exp(-\lambda\,|\tau_i - \tau_j|)$, $\lambda = 8.0$. Specifically: draw $u_k = Lz_k$ where L is the Cholesky factor of Σ and $z_k \sim \mathcal{N}(0, I)$; then $\varepsilon_{ik} = \mathbb{1}\big[\Phi(u_{ik}) > h_i(t)\big]$.

Followers and non-followers of the same decision are assumed uncorrelated. This is a simplifying assumption that could be relaxed in future work.

Three correlation levels are superposed:

(1) Perfect correlation among followers of the same AI decision k.
(2) Partial inter-decision correlation for followers via the shared ξ_t (governed by α).
(3) Profile-distance-based correlation among independent judgers (governed by λ).

A.7 Deskilling

$$h_i(t{+}1) = h_i(t)\big(1 - \eta_{\mathrm{eff}}\,\bar{s}_i(t)\big)$$

where $\bar{s}_i(t) = (1/K)\sum_k s_{ik}(t)$ is the fraction of decisions in which agent i followed the AI. Deskilling is proportional to actual surrender in that quarter; it does not occur in quarters where the agent exercises independent judgement throughout. This is a conservative simplification: diffuse deskilling through tool dependence alone is not modelled.

Scenario	η_{eff}	**Rationale**
Baseline	0.000	No AI exposure
G_0 (unmanaged)	0.020	Proxy: Budzyń et al. [43], −6pp
G_1 (passive)	0.015	Interpolated
G_2 (active)	0.010	Reduced surrender → reduced deskilling

Structural omission. Surrender probability does not depend on $h_i(t)$. The deskilling→surrender feedback loop is absent from the model. Section 4 names this as a source of underestimation of the tail risk premium.

A.8 Output metrics

Raw quarterly loss:

$$L(t) = \sum_{i=1}^{N}\sum_{k=1}^{K} \varepsilon_{ik}(t)$$

Throughput-adjusted loss: $L_\theta(t) = \theta_g \times L(t)$. The parameter θ scales exposure volume: under faster throughput, the same correlated failure propagates across more outputs before detection.

	Baseline	G_0	G_1	G_2
θ_g	1.00	1.25	1.20	1.10

Correct-decision output: $\text{Output}(t) = \theta_g \times (NK - L(t))$.
Excess pairwise error correlation (proposed KRI):

$$\hat{C}_{\text{excess}}(t) = \frac{1}{|P|} \sum_{(i,j)\in P} \left[\frac{1}{K} \sum_k \varepsilon_{ik}\, \varepsilon_{jk} - f_i \times f_j \right]$$

where $f_i = (1/K) \sum_k \varepsilon_{ik}$. This quantity is the proposed KRI of Section 9: its monotonic rise under G_0 and stability under G_2 constitute the direct simulation signature of the Wood et al. [10] mechanism.

Summary statistics across all $T \times M$ quarter-replications: $E[L]$, $P_{99}[L(t)]$, $P_{99}[L_\theta(t)]$, tail ratio $P_{99}/E[L]$, mean output.

A.9 Analytical reference check

The following calculations should be reproduced before running the full simulation as a consistency check. All use the v5 AI error equation from §A.6.

Crisis quarter ($\pi_t = 0.35$, $\xi_t = 2.2$, $\alpha = 0.70$, G_0, $\text{Beta}(2,2)$):

$$P(\text{AI errs} \mid \text{outside}, \xi_t{=}2.2) = \Phi\left(\frac{\sqrt{0.7} \times 2.2 - \Phi^{-1}(0.55)}{\sqrt{0.3}} \right) = \Phi(3.13) \approx 0.999$$

$$P(\text{AI errs} \mid \text{inside}, \xi_t{=}2.2) = \Phi\left(\frac{\sqrt{0.7} \times 2.2 - \Phi^{-1}(0.92)}{\sqrt{0.3}} \right) = \Phi(0.80) \approx 0.79$$

Average AI error probability: $0.65 \times 0.999 + 0.35 \times 0.79 \approx 0.926$.
Effective surrender rate ($\text{Beta}(2,2)$, β_{conform} at Var_{ref}): ≈ 0.69.
Expected loss: $200 \times 10 \times (0.69 \times 0.926 + 0.31 \times 0.40) \approx 1{,}526$.
Implementation A: 1,538 ($\pm$0.8%). Implementation B: 1,551 ($\pm$1.6%). Both within tolerance.

Normal quarter ($\pi_t = 0.70$, $\xi_t = 0$):

$$P(\text{AI errs} \mid \text{outside}, \xi_t{=}0) = \Phi(-0.23) \approx 0.41$$

$$P(\text{AI errs} \mid \text{inside}, \xi_t{=}0) = \Phi(-2.57) \approx 0.005$$

Average AI error probability: $0.30 \times 0.41 + 0.70 \times 0.005 \approx 0.127$.
$E[L \mid \text{normal}] \approx 200 \times 10 \times (0.69 \times 0.127 + 0.31 \times 0.40) \approx 424$.
Both implementations: $\approx$ 424–427 ($\pm$1%).

Baseline (no AI): $E[L] = 200 \times 10 \times 0.40 = 800$. Simulation: ≈ 803 ($\pm$0.4%).

Decision rule: if an implementation does not produce ≈ 1,526 (crisis) / ≈ 424 (normal) / ≈ 800 (baseline) for these three cases, there is a bug. Do not proceed to full simulation runs.

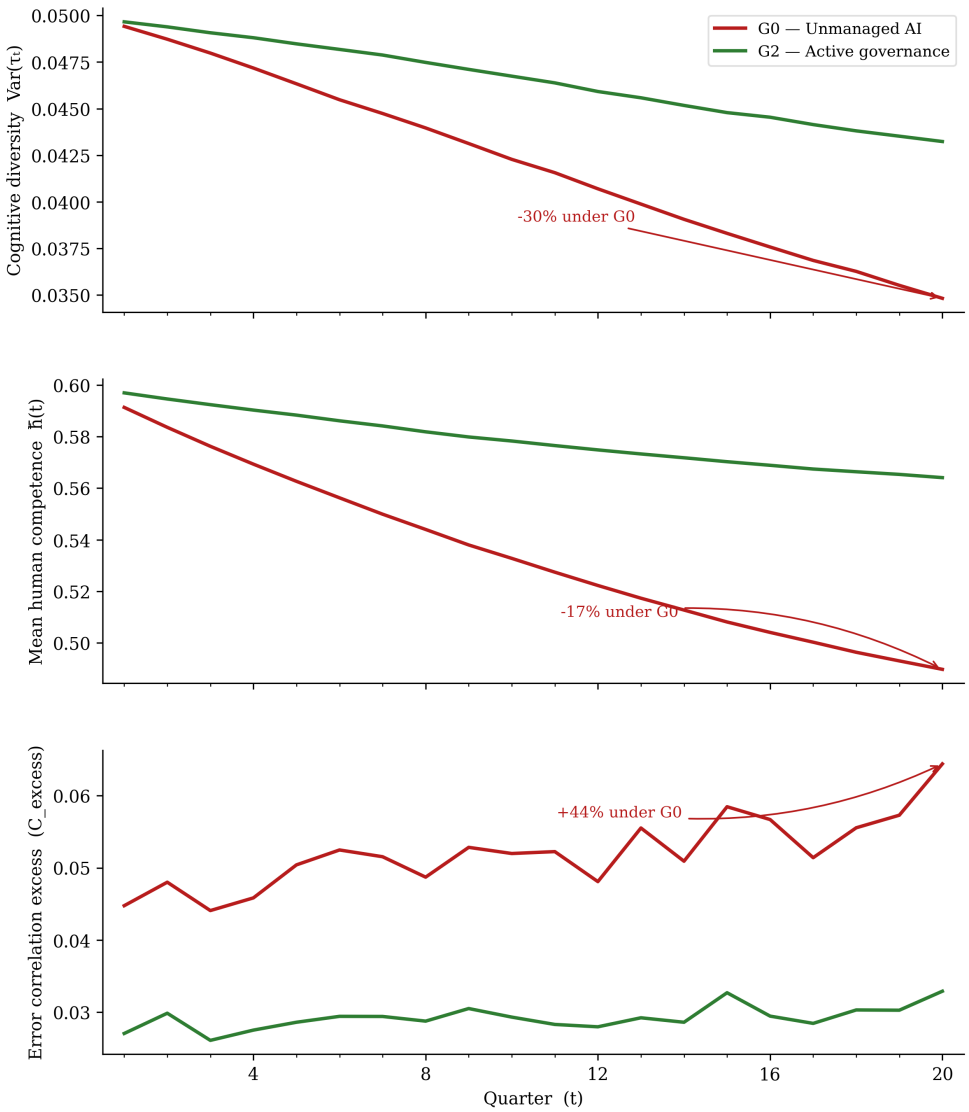

Figure A1. Three channel dynamics under G_0 (red) and G_2 (green) over 20 quarters. *Top*: cognitive diversity $\mathrm{Var}(\tau_t)$. *Middle*: mean capability $\bar{h}(t)$. *Bottom*: pairwise error correlation excess $\hat{C}_{\mathrm{excess}}(t)$. Under G_0, diversity and capability decline monotonically while correlation rises. Under G_2, all three are preserved. *V5 coupled model, under model assumptions.*

B Calibration tables

This appendix documents all 43 parameters[2] of the v5 coupled model. §B.1 reports the central-case values. §B.2 lists profile-specific overrides for P1–P8. §B.3 identifies the five most fragile parameters and their sensitivity ranges.

For each parameter: symbol, description, central value, sensitivity range (where varied), calibration method, epistemic status (one to five stars), and primary source.

[2] 42 original parameters plus ω (D→H trajectory coupling), added in calibration round 2.

B.1 Main parameter table

Symbol	Description	Value	Range	Method	Status	Source
Structural parameters						
$\mathtt{N}$	Number of agents	200	—	Architectural choice	Stylisation	—
$\mathtt{T}$	Horizon (quarters)	20	—	BLS tenure data	Empirical	BLS JOLTS
$\mathtt{K}$	Decisions/agent/quarter	10	—	Architectural choice	Stylisation	—
$\mathtt{M}$	Replications	200	500 (Ex. 1)	Convergence test	Stylisation	—
δ	Quarterly separation rate	0.04	—	BLS JOLTS (prof. services)	⋆⋆⋆⋆⋆	BLS
D-channel (screening)						
$A_{\max}$	Adoption ceiling	0.80	—	SecondTalent/SHRM 2025	Industry	SecondTalent
κ	Adoption growth rate	0.15	—	Exponential fit	Stylisation	—
$\gamma_{\mathtt{eff}}$	Screening intensity (G_0)	5.0	$\{1.5, 3.5, 5.0\}$	Wilson & Caliskan 85.1%	Proxy ⋆⋆⋆⋆	[3]
H-channel (surrender and deskilling)						
p_0	Baseline surrender (G_0)	0.80	$\{0.55, 0.70, 0.80\}$	Shaw & Nave 79.8%	Preprint ⋆⋆⋆⋆	[4]
$\beta_{\mathtt{surr}}$	Surrender sensitivity	0.60	$\{0.4, 0.6, 0.8\}$	Shaw & Nave range	Proxy ⋆⋆⋆	[4]
$\beta_{\mathtt{conform}}$	Conformism coefficient	0.30	$[0.15, 0.35]$	Asch; Lorenz et al.	Proxy ⋆⋆⋆	[51]
$\mathtt{Var}_{\mathtt{ref}}$	Reference variance	0.050	—	$\mathrm{Beta}(2,2)$ exact	Exact	—
λ	Copula decay parameter	8.0	—	Keck & Tang near-zero ρ	Stylisation ⋆⋆⋆	[6][82]
AI quality						

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Symbol	Description	Value	Range	Method	Status	Source
$q_{\texttt{in}}(t)$	AI accuracy (inside)	$0.92+0.002t$	—	Dell'Acqua +40% quality	⋆⋆⋆⋆⋆	[5]
$q_{\texttt{out}}$	AI accuracy (outside)	0.55	—	Dell'Acqua −19pp	⋆⋆⋆⋆⋆	[5][83]
Environment and correlation						
$E[\pi]$	Inside-frontier fraction	0.70	$[0.45, 0.85]$	Profile-specific	Stylisation	—
α	Stack concentration	0.70	$\{0.30, 0.50, 0.70, 1.00\}$	Kim et al. 60% inter-model	Indirect ⋆⋆⋆	[18][84]
η	Deskilling rate (/quarter)	0.02	$\{0.01, 0.02, 0.04\}$	Budzyń et al. −6pp	Clinical proxy ⋆⋆⋆⋆	[43]
Crisis regime (v5 coupling)						
$p_{\texttt{crisis}}$	Crisis probability	0.08	$\{0.05, 0.08, 0.12\}$	Flyvbjerg et al. 8–12%	Anchor ⋆⋆⋆	[64]
$\mu_{\texttt{crisis}}$	Crisis factor mean	2.2	$\{1.5, 2.0, 2.2, 2.5\}$	Grid search (<Basel 3.09)	Stylisation ⋆⋆	—
σ_n	Normal-regime factor SD	0.25	—	Grid search	Stylisation ⋆⋆	—
σ_c	Crisis-regime factor SD	0.30	—	Grid search	Stylisation ⋆	—
$c_{\texttt{normal}}$	Normal Beta concentration	60	—	Low-default analogy	Stylisation	—
$E[\pi_{\texttt{crisis}}]$	Outside fraction in crisis	0.35	—	Dell'Acqua stress proxy	Proxy ⋆⋆⋆⋆	[5]
Throughput						
θ_{G_0}	Throughput (G_0)	1.25	—	Dell'Acqua +25.1% speed	⋆⋆⋆⋆⋆	[5]
θ_{G_1}	Throughput (G_1)	1.20	—	Interpolated	Stylisation	—

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Symbol	Description	Value	Range	Method	Status	Source
θ_{G_2}	Throughput (G_2)	1.10	—	Governance cost estimate	Stylisation	—
Inter-firm and trajectory coupling						
$\kappa_{\texttt{vendor}}$	Vendor correlation	+4%	—	V6 simulation	Output	—
ω	D→H coupling	0.05	{0.02, 0.05, 0.10}	Grid search	Active gap $\star$	—

B.2 Profile-specific overrides (P1–P8)

The values below override the central parameters from §B.1 for each archetype simulation. Where no override is marked, the central value applies. City labels are shorthand for documented labour market structures, not predictions about specific organisations.

† = override from central value. $*$ = expertise-gradient override (p_0 reduced for top-tier profiles following Dratsch et al. [85]).

Profile	$E[\pi]$	α	**Beta**(a, a)	$p_0^{G_0}$	$p_0^{G_2}$	η	**Source**
P1 Frankfurt (Big Four)	0.50	0.70†	3.5	0.75$*$	0.50	0.02	[19]
P2 London (Inv. bank)	0.55	0.45†	1.5†	0.70$*$	0.42	0.02	[77]
P3 Paris (Strategy)	0.55	0.90†	4.0†	0.70$*$	0.42	0.02	[86][87]
P4 Brussels (Legal)	0.45†	0.90†	3.0†	0.80	0.55	0.01†	[88]
P5 SF (Startup)	0.70	0.60†	2.5†	0.80	0.70	0.03†	[70]
P6 Singapore (Creative)	0.85†	0.30†	1.5†	0.80	0.55	0.01†	[20]
P7 Bangalore (Back-office)	0.75†	0.70	2.0†	0.85†	0.78	0.04†	[89][90][91]
P8 Seoul (Central admin.)	0.60	0.95†	4.5†	0.80	0.55	0.01†	[89]

Table A2. Profile-specific parameter overrides. All other parameters take central values from §B.1. $E[\pi]$ is the inside-frontier fraction: $E[\pi] = 0.85$ means 85% of decisions fall inside the AI's competence frontier.

B.3 Five fragile parameters

The following parameters have the largest impact on P_{99} and the weakest empirical anchoring. Sensitivity analysis (Appendix C) varies each across its stated range.

α **(stack concentration).** $\{0.30, 0.50, 0.70, 1.00\}$. Dominant lever. Reducing from $\alpha = 0.90$ to $\alpha = 0.30$ cuts the tail risk premium by approximately 15%. In the v5 coupled model, the ratio $P_{99}(\alpha{=}0.90)$ / $P_{99}(\alpha{=}0.30) \approx 1.15$: the crisis coupling ($\mu_{\text{crisis}} = 2.2$) saturates crisis-quarter errors across all α levels, compressing the marginal effect relative to earlier model versions. Ablation result, not econometric estimate.

$p_{\textbf{crisis}}$ **(crisis frequency).** $\{0.05, 0.08, 0.12\}$. Second most sensitive parameter after α. Direction of the paradox is robust across the full range; magnitude varies.

$\mu_{\textbf{crisis}}$ **(crisis severity).** $\{1.50, 2.00, 2.20, 2.50\}$. Set by grid search across approximately 30 combinations. Conservative relative to Basel II stress scenario ($\Phi^{-1}(0.999) = 3.09$).

$\beta_{\textbf{conform}}$ **(conformism).** $[0.15, 0.35]$. Maximum $\pm 1.5\%$ deviation in P_{99} within this range. "Non-critical" refers to low impact on the tail metric, not to the theoretical importance of the conformism mechanism.

ω **(D→H trajectory coupling).** $\{0.02, 0.05, 0.10\}$. Active gap. Direction is robust: lower diversity raises conformism pressure via β_{conform} regardless of ω. Magnitude is uncertain. See Section 5.

C Sensitivity analysis

This appendix reports single-parameter sensitivity for the five parameters with the largest effect on model results: α (stack concentration), μ_{crisis} (crisis severity), η (deskilling rate), p_{crisis} (crisis frequency), and β_{conform} (conformism coefficient). A sixth parameter, ω (D→H trajectory

coupling), is treated separately in §C.4 as an exploratory extension. Each of the five main parameters is varied across three levels while all others are held at central values.

Two questions are addressed: (1) whether the direction of the mean-tail paradox holds across the full tested range, and (2) how much the magnitude of the tail risk premium varies. The outcome metric throughout is $P_{99} \times \theta$ under G_0. Figure A2 displays the tornado chart.

C.1 Sensitivity summary

Table A3 reports $P_{99} \times \theta$ (G_0) at the low, central, and high level of each parameter. The Δ column gives the absolute and relative spread. The β_{conform} range is formally tested over $[0.15, 0.35]$, the anchored range from Appendix B. An exploratory extension to 0.40 changes $P_{99} \times \theta$ by at most 6 additional points and is not included.

Parameter	**Low**	**Central**	**High**	**Δ range**
α (stack conc.)	1,909	2,135	2,229	319 pts (15%)
μ_{crisis} (severity)	2,016	2,135	2,178	162 pts (8%)
η (deskilling)	2,122	2,135	2,180	58 pts (3%)
p_{crisis} (frequency)	2,114	2,135	2,170	56 pts (3%)
β_{conform} (conformism)	2,111	2,135	2,141	30 pts (1.4%)

Table A3. $P_{99} \times \theta$ (G_0) at low, central, and high parameter levels. V5 coupled model, $M = 200$ replications, $\pm 2\%$ convergence confirmed. Tornado chart: Figure A2.

C.2 Robustness of the mean-tail paradox

The central claim, that AI adoption simultaneously reduces expected loss and increases tail risk, holds at every tested parameter level. Table A4 reports the mean-tail split for each variation. The paradox does not reverse under any tested combination. What varies is the magnitude: the P_{99} increase above baseline ranges from +39% (at $\alpha = 0.30$) to +61% (at $\alpha = 1.00$), with the central estimate at +56%.

Parameter level	**$E[L]$ vs baseline**	**P_{99} vs baseline**	**Paradox?**
$\alpha = 0.30$ (low)	-35%	$+39\%$	Yes
$\alpha = 0.70$ (central)	-38%	$+56\%$	Yes
$\alpha = 1.00$ (high)	-40%	$+61\%$	Yes
$\mu_{\text{crisis}} = 1.50$	-39%	$+45\%$	Yes
$\mu_{\text{crisis}} = 2.50$	-37%	$+57\%$	Yes
$p_{\text{crisis}} = 0.05$	-41%	$+54\%$	Yes
$p_{\text{crisis}} = 0.12$	-33%	$+58\%$	Yes
$\eta = 0.01$ (low)	-37%	$+54\%$	Yes
$\eta = 0.04$ (high)	-38%	$+57\%$	Yes
$\beta_{\text{conform}} = 0.15$	-38%	$+55\%$	Yes
$\beta_{\text{conform}} = 0.35$	-38%	$+56\%$	Yes

Table A4. Mean-tail split across all parameter variations. Changes are relative to no-AI baseline (P_{99} brut, not throughput-adjusted). V5 coupled model, $M = 200$ replications.

C.3 Parameter-level discussion

α **(stack concentration).** Dominant lever, approximately 2:1 over the next parameter. Reducing α from 0.70 to 0.30 reduces $P_{99} \times \theta$ from 2,135 to 1,909 and the P_{99} brut premium from +56% to +39% above baseline. At $\alpha = 1.00$, $P_{99} \times \theta$ reaches 2,229.

This dominance makes stack concentration a natural regulatory target. Whether that target is operationalisable as a mandated standard depends on how governance frameworks define and audit internal AI workflow architecture, a question beyond the model's scope.

μ_{crisis} **(crisis severity).** Second most sensitive parameter (range 162 points). The central value 2.2 is conservative relative to the Basel II stress scenario ($\Phi^{-1}(0.999) = 3.09$). At $\mu_{\text{crisis}} = 1.50$, $P_{99} \times \theta$ falls to 2,016 (+45% premium). At 2.50 it reaches 2,178. This parameter governs how severely the AI is miscalibrated when the environment shifts. It is a calibrated stylisation; the sensitivity range is reported precisely because the central value lacks direct empirical grounding.

p_{crisis} **(crisis frequency).** Produces an asymmetric pattern. As crisis frequency rises from 0.05 to 0.12, the mean gain from AI adoption shrinks (-41% to -33% on $E[L]$) while the tail premium grows (+54% to +58%). More crisis quarters means more quarters where the AI is unreliable, eroding the average benefit while amplifying the tail. The central calibration $p_{\text{crisis}} = 0.08$ may be conservative for sectors facing more frequent structural shocks than the IT project base rate anchoring it [64].

η **(deskilling rate).** Small direct effect on $P_{99} \times \theta$ in the single-parameter protocol (58 points, $\sim$3%), because deskilling accumulates slowly relative to the crisis-driven tail. Its importance is primarily longitudinal: over 20 quarters, $\bar{h}$ declines by 17.2% under G_0 (from 0.591 to 0.490), the deskilling result underpinning the irreversibility argument of Section 8. The sensitivity range spans the proxy transfer from clinical medicine [43, 92] to knowledge work. Direction is robust throughout.

β_{conform} **(conformism).** Non-critical within the tested range $[0.15, 0.35]$: $P_{99} \times \theta$ varies from 2,111 to 2,141, a spread under 1.5%. β_{conform} is theoretically important as the formal expression of the D→H coupling, but its magnitude does not drive the quantitative results within the anchored range.

C.4 Exploratory extension: ω

The parameter ω is not part of the final calibrated model. In Appendix A, the D→H coupling is carried entirely by β_{conform}: as $\text{Var}(\tau)$ falls, the conformism mechanism raises p_0^{eff}, amplifying surrender. ω represents a second, direct coupling pathway: the possibility that diversity compression raises surrender rates through channels not captured by β_{conform} alone (accelerated information cascades, reduced access to dissenting colleagues).

ω is explored as a sensitivity extension rather than a calibrated parameter because no direct empirical estimate exists. Preliminary grid search over $\{0.02, 0.05, 0.10\}$ as an additive amplifier to the β_{conform} pathway produces $P_{99} \times \theta$ (G_0) ranging from approximately 2,100 to 2,220, a spread comparable to the μ_{crisis} sensitivity. The direction of the paradox holds throughout.

The absence of ω from the final model is a structural omission whose direction is known: including it would increase the estimated tail risk premium, not reduce it. Section 9 identifies measuring the direct D→H coupling as the primary empirical priority.

C.5 Note on parameter interactions

All tests in §C.1–§C.3 are single-parameter variations. The model contains at least two interaction effects not captured by this protocol.

First, α and β_{conform} interact multiplicatively. A homogeneous organisation (amplified β_{conform} effect) that also uses a centralised AI workflow (high α) faces compound amplification exceeding the sum of individual effects. This interaction is the structural source of the Paris–London contrast in Section 7.

Second, p_{crisis} and μ_{crisis} jointly determine crisis severity. Holding one fixed while varying the other understates the joint tail exposure. A grid search over $(p_{\text{crisis}}, \mu_{\text{crisis}})$ pairs is on the empirical agenda.

Conclusion. Across all tested single-parameter variations, the mean-tail paradox holds without exception. $P_{99} \times \theta$ (G_0) ranges from +39% to +61% above baseline, with +56% as the central estimate. α is the dominant lever. β_{conform} is the least quantitatively sensitive parameter within its anchored range. ω remains outside the calibrated model; its inclusion would increase the tail risk premium.

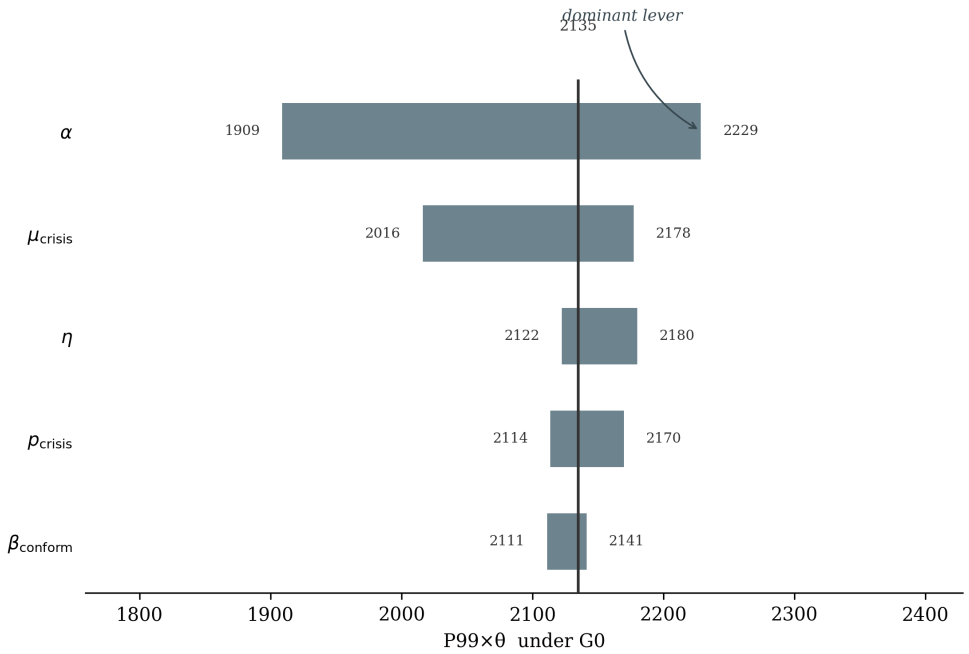

Figure A2. Single-parameter sensitivity (tornado chart). $P_{99} \times \theta$ under G_0 across the tested range of five parameters. Central value: 2,135. α dominates (319 pts, 15%); μ_{crisis} is second (162 pts, 8%); η, p_{crisis}, β_{conform} have small direct effects within tested ranges. *V5 coupled model, single-parameter variation.*

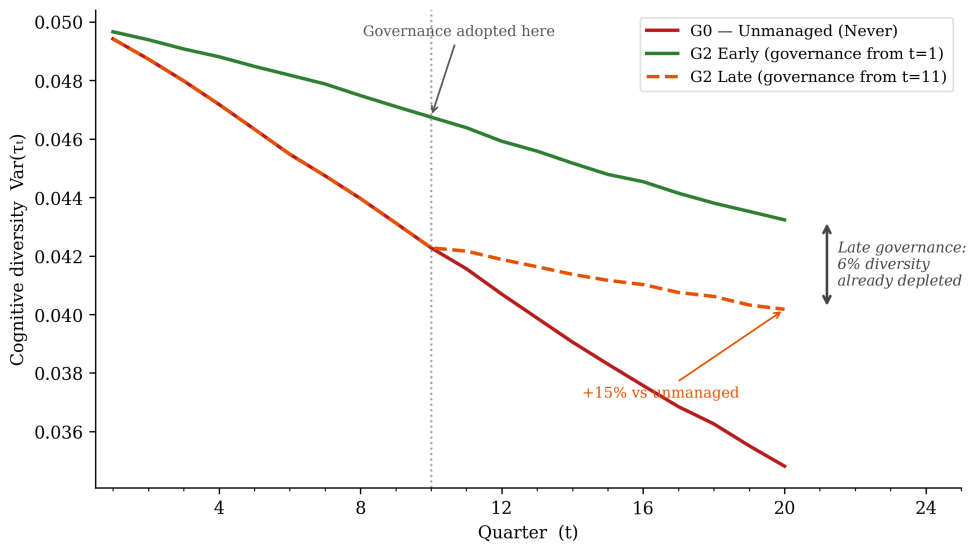

Figure A3. Partial irreversibility of cognitive diversity under early versus late governance. G_0 (red): continuous unmanaged adoption, $\mathrm{Var}(\tau)$ declines monotonically. G_2 Early (green): governance from $t = 1$. G_2 Late (orange, dashed): governance adopted at $t = 10$ (≈2.5 years). Late governance arrests the decline but cannot restore the 6% shortfall in $\mathrm{Var}(\tau)$ already depleted. *V4 verification, v5 coupled model, under model assumptions.*

D Ablation study

This appendix reports $P_{99} \times \theta$ and derived metrics for all eight organisational archetypes across all four governance scenarios. The goal is to make the profile-level results fully inspectable, separating what varies by structural configuration from what is driven by governance choice. Figure A5 displays the heatmap. Figure A6 displays the scaffold benefit bar chart.

Profiles are sorted by absolute $P_{99} \times \theta$ under G_0. Absolute level is *not* the primary reading. The most analytically important dimensions are: (1) the premium of G_0 relative to the no-AI baseline, which captures structural amplification; (2) the scaffold benefit, which measures governance efficiency rather than outcome level; and (3) the tail ratio, which is size-independent. A profile ranked low by absolute $P_{99} \times \theta$ may carry the most severe relative risk trajectory. The Paris archetype is the clearest example.

D.1 Main ablation table

Values are $P_{99} \times \theta$ (throughput-adjusted 99th-percentile loss) under each governance scenario. *Baseline* is the no-AI reference. *Premium* G_0 is the percentage increase from Baseline to G_0. *Tail ratio* is P_{99} brut $/E[L]$ under G_0. *Scaffold benefit* is the build_summary efficiency metric defined in §D.3.

D.2 Key contrasts

Paris–London (P3 vs P2). Same sector ($E[\pi] = 0.55$), same talent range ($h \in [0.65, 0.95]$), same baseline surrender rate ($p_0 = 0.70$). The only structural differences are stack concentration ($\alpha = 0.90$ vs 0.45) and cognitive profile (Beta$(4, 4)$ vs Beta$(1.5, 1.5)$). Under these parameters, $P_{99} \times \theta$ under G_0 is 2,041 vs 1,668, a gap of 22%. This is the most tightly controlled contrast in the table: the two parameters that differ (α and Beta(a, a)) are those the paper argues are structurally determined by labour market conditions rather than managerial choice.

Seoul–Singapore (P8 vs P6). Extreme ends of the stack concentration range ($\alpha = 0.95$ vs 0.30) combined with opposite diversity profiles (Beta$(4.5, 4.5)$ vs Beta$(1.5, 1.5)$). Result: $P_{99} \times \theta$ of 2,318 vs 1,845, roughly 25% more tail risk for the Seoul archetype ($\times 1.26$). This contrast illustrates the upper bound of the structural spread; it does not constitute a prediction about organisations in Seoul or Singapore.

Elite premium paradox (P3 Paris). $E[L]_{G_0} = 481$, the lowest mean loss among all profiles, because top-tier talent exercises high-quality independent judgement in normal quarters. Yet $P_{99} \times \theta_{G_0} = 2{,}041$, a +198% premium above baseline, the highest proportional tail risk in the set. Paris makes fewer errors on average and far more in the worst quarters. This inversion is the numerical expression of the top-tier twist (Section 7): high capability is what AI erodes most efficiently, via the combination of high outside-frontier exposure and a centralised homogeneous stack.

Passive governance (G_1). The mean G_1 tail reduction across all profiles is 12.1%, approximately one-fifth of the reduction achievable under G_2. For P5 SF (7.7%) and P7 Bangalore (6.3%), passive governance is nearly ineffective. For P2 London (20.2%) and P3 Paris (19.2%), it delivers meaningful but partial protection. G_1 is not a substitute for G_2; it is at best a transitional posture. Figure A4 plots all eight archetypes on the output–risk plane.

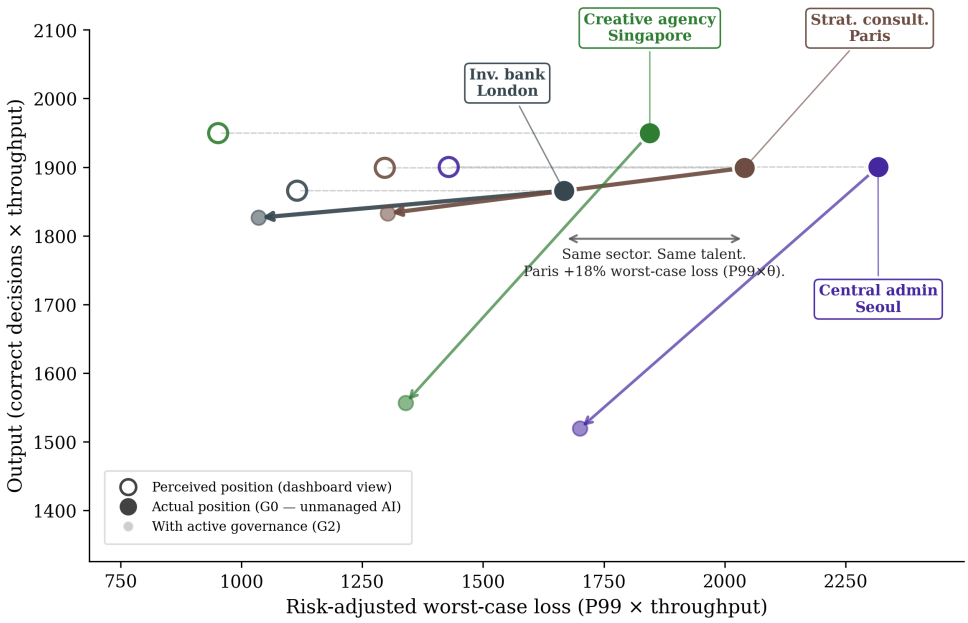

Figure A4. Eight organisational profiles under G_0: output relative to no-AI baseline (horizontal) versus $P_{99} \times \theta$ (vertical). London and Paris share sector and talent ($E[\pi] = 0.55$, $h \in [0.65, 0.95]$); their separation ($\alpha = 0.45$ vs 0.90) produces a 22% gap in tail risk. Paris carries the highest proportional premium (+198%), the lowest expected loss, and the steepest scaffold benefit: the numerical expression of the top-tier twist. The frontier is a governance-exposure map, not a production frontier. *V5 coupled model, $M = 200$ replications, under model assumptions.*

D.3 The scaffold benefit metric

The scaffold benefit is the build_summary efficiency metric. It measures whether G_2 achieves a better output-to-risk trade-off than G_0:

$$\text{scaffold_benefit} = \frac{\text{ratio}_{G_2} - \text{ratio}_{G_0}}{\text{ratio}_{G_0}} \qquad \text{where} \quad \text{ratio}_G = \frac{\text{Output}_G / \text{Output}_{\text{base}} - 1}{P_{99,G} / P_{99,\text{base}} - 1}$$

A positive value means G_2 achieves a better output-to-risk ratio than G_0: governance is efficient. A negative value means governance costs more in throughput than it saves in tail risk, relative to the unmanaged baseline. The metric is neutral on whether reducing tail risk at the cost of output is worthwhile; that is a governance question, not a model output.

Critical distinction. A negative scaffold benefit does not mean G_2 fails to reduce tail risk. For P5 SF, G_2 reduces $P_{99} \times \theta$ from 2,154 to 1,668, a 22.5% reduction. The scaffold benefit is negative (-36%) because the throughput cost (θ falls from 1.25 to 1.10) makes the trade-off unfavourable under this profile's parameters. Governance reduces risk for all eight profiles. Its *efficiency*, risk reduction per unit of output foregone, is what is regressive.

D.4 Contextual anchors

Ecological diversity. Huang et al. (*Science*, 2018) document that mixtures of 16 plant species produce more than twice the above-ground carbon biomass and approximately 80% more productivity than monocultures under identical conditions. The analogy is structural: both cases involve a shared resource whose resilience depends on variance of approach, not average quality. It does not imply identical mechanisms; it contextualises the direction of the ablation results.

Financial concentration. Moeller, Schlingemann, and Stulz (*Journal of Finance*, 2005) document approximately $240 billion in value destruction by acquiring firms during the 1998–2001 M&A wave, a period characterised by herding behaviour and concentrated analytical frameworks. The specific concentration ratios reported in that paper could not be independently verified in the course of this project and are not reproduced here. The qualitative finding, synchronised decision-making under homogeneous frameworks produces concentrated losses, is directionally consistent with the ablation results.

All results in this appendix are model outputs under stated assumptions. They quantify variation in tail risk attributable to structural dimensions. They are not empirical estimates of the risk faced by any specific organisation.

	Baseline	G0	G1	G2	Scaffold benefit
Big Four audit (Frankfurt)	924	2130	1856	1446	+3%
Investment bank (London)	635	1475	1199	961	+156%
Strategy consulting (Paris)	686	1994	1603	1272	+97%
Corporate legal (Brussels)	1121	2231	1987	1585	-4%
Tech startup (SF)	958	1895	1780	1455	-34%
Creative agency (Singapore)	1084	1454	1332	1133	-122%
Back-office (Bangalore)	1407	2120	1992	1669	-16%
Central admin (Seoul/Tokyo/Paris)	1203	2312	2068	1669	+0%

Figure A5. Ablation table: $P_{99} \times \theta$ by profile and governance scenario. Dark red = high tail risk (G_0); green = lower risk (G_2). Rightmost column: scaffold benefit (green = governance efficient; red = governance penalises efficiency). Profiles sorted by $P_{99} \times \theta_{G_0}$ descending. *V5 coupled model, $M = 200$, under model assumptions.*

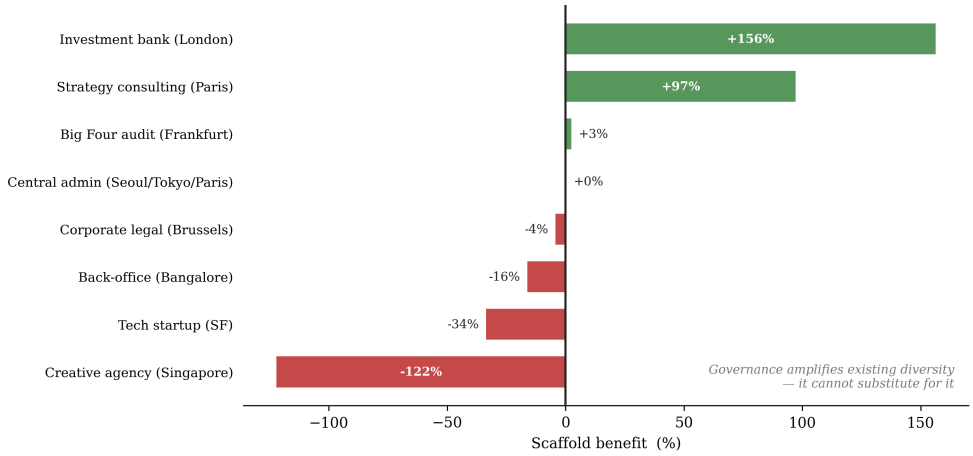

Figure A6. Scaffold benefit by archetype. Positive (green): G_2 improves the output-to-risk trade-off. Negative (red): G_2 imposes an efficiency loss. A negative value does not mean G_2 fails to reduce tail risk; it means the throughput cost outweighs the risk reduction for that profile. *V5 coupled model,* $M = 200$*, under model assumptions.*

E Bibliographic notes

This appendix documents the 13 sources that carry particular weight in the argument or that require transparency about their limitations. For each source: bibliographic reference, epistemic status, role in the essay, limits of the evidence, and verification notes. Sources are grouped by function.

Four base epistemic statuses are used: *Peer-reviewed* (journal or conference with referee process); *Preprint* (posted but not yet peer-reviewed; preregistered where noted); *Industry source* (report or white paper); *Working paper* (circulated but not refereed). Within peer-reviewed sources, the status bracket is enriched locally by evidence type (field experiment, IV identification, clinical RCT, etc.) to give the reader a precise reading of what the source establishes. All enriched labels fall under *Peer-reviewed*.

E.1 Theoretical core

[10] — Wood et al. (2023), JMLR.
Peer-reviewed (journal)

Role. Central theoretical anchor for the mean-tail paradox. The decomposition $\mathrm{Var}(L) \propto \bar{p}(1-\bar{p})(1+(NK-1)\bar{\rho})$ shows that when pairwise error correlation $\bar{\rho}$ rises, aggregate variance grows even if mean error $\bar{p}$ falls. The model uses this as its mechanistic claim: AI reduces $\bar{p}$ while $\bar{\rho}$ rises, and the tail follows. The earlier [9] ambiguity decomposition is the special case; Wood et al. generalise it.

Limits. The decomposition is exact for squared and cross-entropy loss in ensemble settings. The transfer to organisational decision aggregation requires that individual errors behave as ensemble predictors, an assumption that holds structurally in the model but has not been validated in real organisations.

Verification. arXiv:2301.03962. Cited for the formal decomposition, not as an application to AI-assisted organisations. That application is our framing.

[18] — Kim, Garg, Peng & Garg (2025), ICML.
Peer-reviewed (conference)

Role. Primary empirical anchor for α. The study finds approximately 60% agreement between two LLMs when both give an incorrect answer, across 350+ models in the HELM benchmark. This establishes that AI errors are not independent across models, and by extrapolation, not independent across decisions within the same model. The model sets $\alpha = 0.70$ for a centralised single-model stack, higher than the 0.35–0.45 inter-model figure, on the reasoning that intra-model correlation exceeds inter-model correlation.

Limits. The 60% figure is inter-model error agreement, not intra-model inter-decision correlation (what α directly captures). The extrapolation is a modelling assumption, not a measured quantity. This gap is documented in §B.3. Citation must always use the full four-author list: Kim, Garg, Peng & Garg.

Verification. arXiv:2506.07962. Four authors verified. Venue: ICML 2025, PMLR 267.

[6] — Keck & Tang (2020), Psychological Science.
Peer-reviewed (journal)

Role. Central empirical anchor for the D→H bridge. Groups with cognitive-process diversity produce near-zero pairwise error correlation; homogeneous groups produce strongly correlated errors. The copula parameter $\lambda = 8$ is calibrated to reproduce the near-zero correlation result for maximally distant cognitive types.

Limits. Laboratory result, not an organisational field study. The transfer requires: (1) cognitive profile diversity maps onto decision-process diversity; (2) the correlation structure holds under organisational conditions with AI assistance. Neither transfer has been validated. This is the primary empirical gap (Section 9).

Verification. DOI: 10.1177/0956797620941840. Cited for the direction of the correlation–diversity relationship. The λ calibration uses this result as a target, not as a direct estimate.

E.2 Key empirical anchors

[4] — Shaw & Nave (2026), PsyArXiv.
Preprint (preregistered)

Role. Primary anchor for $p_0 = 0.80$ under G_0. Shaw & Nave report that 79.8% of participants follow an AI recommendation even when it is incorrect ($N = 1{,}372$, 9,593 trials, GPT-4o, CRT items). Study 3 override rate under monetary incentives (42.3%) anchors β_{surr} and informs $p_0^{G_2}$.

Limits. Preprint. CRT-style tasks administered online may not generalise to professional knowledge-work decisions. The 79.8% is the ceiling estimate; effective surrender under $\text{Beta}(2, 2)$ is approximately 0.69 after the β_{surr} penalty. A preprint critique by Stanković (2026, arXiv) questions the ecological validity of the CRT transfer; noted here without resolution, as the main direction is convergent with [37] and [93].

Verification. DOI: 10.31234/osf.io/yk25n. Preregistered. Cited as preprint throughout the body. The Stanković critique has not been peer-reviewed.

[5] — Dell'Acqua et al. (2026), Organization Science.
Peer-reviewed (journal, field experiment)

Role. Primary anchor for $q_{\text{in}} = 0.92$, $q_{\text{out}} = 0.55$, and $\theta = 1.25$. Field experiment with 758 BCG consultants. Inside the AI frontier: +40% quality, +25.1% speed. Outside: −19pp vs unassisted controls. Skill compression: bottom performers +43%, top performers +17%.

Limits. Our framing diverges from the authors'. Dell'Acqua et al. frame the outside-frontier result as a risk to manage. We frame the skill compression as the mechanism through which AI erodes the cognitive diversity premium, an interpretation more aggressive than their central claim. Readers should consult the original for the authors' framing. Nine-author paper: cite as Dell'Acqua et al. in text, full list in bibliography.

Verification. DOI: 10.1287/orsc.2025.21838. Preregistered RCT with 758 participants. The skill compression reading is identified as our analytical extension.

[43] — Budzyń et al. (2025), Lancet Gastroenterology & Hepatology.
Peer-reviewed (journal, clinical RCT)

Role. Direction anchor for $\eta = 0.02$/quarter. Endoscopists in the AI-assisted group show a -6pp decline in adenoma detection rate after withdrawal of AI assistance. Direct causal measurement of deskilling from AI reliance in a clinical setting.

Limits. Clinical medicine result. The -6pp "over a period" translates to $\eta \approx 0.02$/quarter by proxy, not by direct calibration. The 21-author paper requires "Budzyń et al." in text with full citation in bibliography. η is a direction anchor; the exact magnitude in knowledge-work settings is unvalidated.

Verification. DOI: 10.1016/S2468-1253(25)00133-5. Cited for direction only.

[64] — Flyvbjerg et al. (2022), JMIS.
Peer-reviewed (journal, two independent studies)

Role. Order-of-magnitude anchor for $p_{\text{crisis}} = 0.08$. Catastrophic IT project failures at 8–12% frequency across two independent datasets. The 2022 JMIS paper (six authors: Flyvbjerg, Budzier, Lee, Krawczyk, Vecci, Clegg) documents the base rate. A companion study shows IT projects have Pareto tail exponent $\alpha_{\text{Pareto}} \leq 1$, implying infinite variance in principle, which justifies regime-switching rather than continuous distributions [65].

Limits. Failure rates are measured at the project level, not at the quarterly decision level used in the model. The translation to "crisis quarter frequency" is a modelling choice. The 8% figure is an order-of-magnitude anchor; sensitivity range $\{0.05, 0.08, 0.12\}$ explores its influence.

Verification. JMIS 2022: DOI 10.1080/07421222.2022.2063518. Six-author paper. Two independent datasets with convergent conclusions strengthen the anchor despite the level-of-analysis gap.

E.3 Preprints requiring explicit status

[12] — Holzner, Maier & Feuerriegel (2025), arXiv.
Preprint

Role. Anchor for the divergent individual/collective AI effects: $g = +0.27$ for individual creativity, $g = -0.86$ for collective diversity, across 379 conditions. Quantifies the within-individual vs. across-individual asymmetry central to the paper's argument.

Limits. Preprint. Meta-analyses are sensitive to inclusion criteria and publication bias; the $g = -0.86$ collective result is large and should be treated as directional. Cited with explicit preprint flag (!) throughout.

Verification. arXiv:2505.17241. Cited for direction only.

[94] — Kosmyna et al. (2025), arXiv:2506.08872.
Preprint (methodological critique exists)

Role. Supporting context for the cognitive offloading narrative. EEG study reports −55% dDTF connectivity in participants using ChatGPT compared to unaided recall.

Limits. Preprint. A formal methodological critique (arXiv:2506.11676) challenges the dDTF interpretation. Neither preprint has been peer-reviewed. This source is **excluded from the main body** by policy and appears here only as context. The −55% figure is not used in any model parameter or argument.

Verification. arXiv:2506.08872. Critique: arXiv:2506.11676. Not in refs.bib.

E.4 Methodological and conceptual sources

[71] — Danielsson & Shin (2002, 2012).
Peer-reviewed (multiple publications)

Role. Source of the Millennium Bridge analogy in Section 8. Pedestrians adapt their gait to bridge oscillations, amplifying rather than damping them. Financial institutions do the same with VaR models. The analogy contextualises the Nash equilibrium argument: competitive pressure generates correlated vulnerability regardless of individual rationality.

Limits. The analogy is structural, not quantitative. It establishes the feedback amplification mechanism but does not calibrate its magnitude. The transfer from civil engineering to financial risk to AI adoption involves two analogical steps.

Verification. Multiple publications, multiple venues. 2002 and 2012 formulations in refs.bib.

[73] — Bernile, Bhagwat & Yonker (2018), J. Financial Economics.
Peer-reviewed (journal, IV identification)

Role. One of the few studies with causal identification on the diversity-to-risk relationship at the organisational level. Using instrumental variables, board-level cognitive diversity reduces firm-level return volatility and improves R&D efficiency. Cited in Section 8.

Limits. Board-level diversity is one dimension, not full-workforce cognitive profile diversity as modelled here. The IV strategy may be debated. The paper defines cognitive diversity through observable board-member characteristics, not through a latent τ profile.

Verification. DOI: 10.1016/j.jfineco.2017.12.009. IV strategy described in the main text.

[66] — Bainbridge (1983), Automatica.
Peer-reviewed (journal)

Role. Conceptual anchor for the θ masking mechanism. Bainbridge observes that systems designed to accelerate work render their own failure modes invisible. The θ parameter formalises this observation: a throughput multiplier that improves observed output while masking tail risk accumulation.

Limits. Theoretical essay based on industrial process control, not an empirical study. Applicability to AI-assisted knowledge work rests on structural analogy. The "masking" framing is our formalisation, not the paper's.

Verification. DOI: 10.1016/0005-1098(83)90046-8. The masking interpretation is our extension; Bainbridge provides the foundational observation.

[67] — Kahneman, Sibony & Sunstein (2021), *Noise*.
Peer-reviewed (book)

Role. Contextualises the -38% $E[L]$ result in Section 7. Kahneman et al. document that human decision noise is enormous: inter-rater variance reaching 55% in insurance underwriting and spanning 5–88% in asylum adjudication.

Limits. The protective-noise framing, that unsynchronised noise has a systemic resilience property through error decorrelation, is our analytical extension, not the authors' thesis. Kahneman et al. argue noise is a flaw to correct. We argue that noise's decorrelated structure has a protective function that AI adoption destroys. The figures (55%, 5–88%) come from studies reported in the book, not from original research by the authors.

Verification. Little, Brown, 2021. Protective-noise framing explicitly identified as our extension in the body (Section 7).

F Reproducibility note

This appendix documents the steps required to reproduce the simulation results. The model was implemented in two separate coding environments using the specification in Appendix A and the same seed protocol. All reported results are from Implementation A; Implementation B serves as the convergence check.

F.1 Code repository

The simulation code is available at:

`[Repository URL -- to be inserted upon submission]`

The repository contains: the full simulation code (Python 3.11, NumPy 1.26), all parameter files for model versions v1–v5, the seed file, output CSVs for all scenarios and profiles, and figure-generation scripts for Exhibits 1–4 and Figures A1–A5.

F.2 Seeds and replication protocol

All simulations use fixed seeds for exact reproducibility:

Component	Value
BASE_SEED	42
SCENARIO_SEEDS	baseline: 1000, G0: 2000, G1: 3000, G2: 4000
PROFILE_OFFSET	profile_id × 100 (P1=100 … P8=800)
seed_i	SCENARIO_SEED + PROFILE_OFFSET + replication_index

Replication index runs from 0 to $M-1$ (0–199 for $M = 200$; 0–499 for $M = 500$). The $M = 500$ runs used for Exhibit 1 extend the same seed sequence from index 0 to 499; they do not use a separate block. The first 200 replications in both $M = 200$ and $M = 500$ runs are therefore identical.

F.3 Two-implementation convergence check

Implementation A. Python 3.11, NumPy 1.26, SciPy 1.11. Primary implementation. All figures and reported numerical values are drawn from this implementation.

Implementation B. GPT-4o code execution environment, equivalent NumPy operations. Coded independently from the same specification (Appendix A), without access to Implementation A's source code.

Both implementations use the same specification, the same seeds, and similar numerical libraries. "Independent" refers to independence of code: no source code was shared. The check demonstrates that two separate codings of the same specification produce quasi-identical outputs, not that all sources of numerical variation have been controlled.

Metric	**Impl. A**	**Impl. B**	Δ
$E[L]_{G_0}$	498.9	502.1	+0.6%
P_{99} brut (G_0)	1,708	1,723	+0.9%
$P_{99} \times \theta$ (G_0)	2,135	2,154	+0.9%
Output (G_0)	1,876	1,872	−0.2%
Tail ratio (G_0)	3.42	3.43	+0.3%
scaffold_benefit P2 London	+146%	+144%	−1.4%
scaffold_benefit P5 SF	−36.0%	−33.1%	+2.9%

Table A5. Two-implementation convergence check. All differences within ±2% on primary metrics and ±3% on derived metrics (scaffold_benefit).

All reported figures in the paper use Implementation A values. The largest divergence is scaffold_benefit P5 SF at 2.9%, within tolerance and consistent in sign and magnitude.

F.4 Model version history

Results use the v5 coupled model (finalised 24 March 2026). Earlier versions are archived in the repository.

v2 → v3. Introduction of the Vasicek single-factor α for inter-decision correlation. λ recalibrated from 3.0 to 8.0 to correct excessive baseline correlation. First version where bimodality emerges: two modes appear in the loss distribution, though the gap between them is not yet sharply separated.

v3 → v4. Addition of β_{conform} (conformism coupling; later adjusted from 0.35 to 0.30 in v5). Expansion from 5 to 8 organisational profiles. Un

v4 → v5. Introduction of the π-ξ regime coupling ($\mu_{\text{crisis}} = 2.2$, $\sigma_n = 0.25$, $\sigma_c = 0.30$); β_{conform} adjusted to 0.30. This coupling produces clean separation into two distinct regimes: the gap between the normal-quarter mode and the crisis-quarter mode becomes near-empty, and 99.7% of crisis-mode observations carry a crisis flag.

Metric	**v4**	**v5**
$E[L]_{G_0}$	525	499
P_{99} brut (G_0)	1,555	1,708
$P_{99} \times \theta$	1,965	2,135

Table A6. Central-case changes v4→v5. Direction of all main results unchanged; tail premium larger in v5 due to crisis coupling. No v4 values should appear in the paper.

F.5 Analytical reference check

The analytical consistency check is documented in §A.9. Before running any simulation, three reference calculations should be reproduced analytically:

Case	**Analytical**	**Impl. A**	**Impl. B**
Crisis (π_t=0.35, ξ_t=2.2)	$\approx$ 1,526	1,538	1,551
Normal (π_t=0.70, ξ_t=0)	$\approx$ 424	424	427
Baseline (no AI)	800	803	803

Table A7. Analytical reference vs. simulation output. All within $\pm$2%. If an implementation does not produce these values, there is a bug.

The full derivation of the analytical values is in §A.9.

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